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Simultaneous stomach-brain electrophysiology reveals dynamic coupling in human sleep

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Abstract

Sleep involves continuous communication between the brain and body, yet the dynamics of peripheral signals during human sleep remain poorly understood. Here we tested whether gastric electrophysiology exhibits infraslow structure and coordinated fluctuations with cortical rhythms indicative of sleep physiology. Simultaneous high-density electroencephalography (EEG) and electrogastrography (EGG) were recorded across sixty participants and three nights. Gastric power was consistently higher during NREM than REM sleep and declined across successive cycles, consistent with stage-dependent autonomic modulation of visceral activity. For the first time, we show that the gastric rhythm itself exhibits intrinsic infraslow amplitude modulations (∼0.007 Hz), which are selectively

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amplified during NREM sleep and temporally aligned with infraslow fluctuations in cortical sigma power, strongest during N3 sleep. Event-locked analyses further revealed transient increases in gastric amplitude following cortical slow wave oscillations, particularly when accompanied by sleep spindles. Across nights, variance in gastric infraslow amplitude predicted subjective sleep quality beyond standard polysomnographic and cardiac measures. Together, these findings position the human stomach as a peripheral oscillator whose infraslow dynamics track thalamocortical activity during sleep and predict subjective sleep quality, extending the interoceptive regulatory loop into the sleeping brain.

Keywords: sleep physiology, brain-body coupling, electrogastrography, interoception

1 Introduction

Across species, sleep supports restoration, memory consolidation, and homeostatic regulation of the body and brain [1]. Traditionally viewed as a state of sensory and motor disconnection [2], sleep is now recognized as an active process that orchestrates communication between central and peripheral physiology. The autonomic nervous system (ANS) forms the primary conduit for this communication, coordinating brain activity with visceral function to maintain internal stability [3]. Sleep stages impose distinct organizational modes on this coupling. In non-rapid eye movement (NREM) sleep, parasympathetic dominance supports deeper synchronization between central and autonomic processes, whereas rapid eye movement (REM) sleep is marked by cortical desynchronization, muscle atonia, and autonomic changes with phasic surges in heart rate, respiration, and sympathetic output [2]. These alternating modes reflect rhythmic shifts between coupling and uncoupling across the brain-body axis.

Within NREM sleep, large-amplitude cortical slow wave oscillations (SWOs) and thalamocortical spindles dominate neural activity, shaping the temporal framework for communication between cortical and subcortical regions [4, 5]. Superimposed on these rhythms, infraslow oscillations (<0.1 Hz) further modulate arousability by synchronizing cortical, thalamic, and brainstem activity [6, 7]. Driven by cyclic changes in locus coeruleus-noradrenergic tone and parasympathetic outflow, these oscillations delineate phases of neural quiescence and transient activation, allowing the brain to preserve sleep stability while remaining responsive to salient cues. Parallel infraslow fluctuations in heart rate [8], respiration [9], and blood glucose [10] indicate a unified central–autonomic control process operating on a scale of tens of seconds. Thus, rather than a passive withdrawal from the environment, NREM sleep represents a state of coordinated brain-body regulation where central and peripheral rhythms interact to maintain internal balance and adaptive responsiveness.

However, the functional relevance of this coordination remains uncertain. It may sustain homeostatic regulation, support restorative processes such as metabolic recovery, or preserve readiness for sudden arousal. Identifying visceral systems that maintain coordinated dynamics with the brain, even during states of reduced neural communication, can reveal how sleep integrates internal and external demands. Among these

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systems, the stomach represents a compelling system for exploring brain-body coupling [3]. It is the most vagally innervated organ in the gastrointestinal tract [11, 12], and its rhythmic contractions, paced by interstitial cells of Cajal, produce a slow electrical wave ($\sim$ 0.05 Hz) measurable with non-invasive electrogastrography (EGG) [13, 14]. Recent studies show these gastric slow waves can synchronize with cortical activity during quiet wakefulness, suggesting that visceral rhythms may scaffold large-scale organization of neural circuits within the brain [15, 16].

Despite this evidence in wakefulness, the contribution of gastric rhythms to sleep physiology remains largely unexplored. Prior gut-sleep studies have focused on non-electrical markers (e.g., microbiota composition [17], gastric acid secretion [18], and bowel sound recordings [19]) or on EGG frequency shifts in gastric disorders [20], rarely assessing direct stomach-cortex coupling in humans. A handful of animal studies suggest visceral inputs can reach and modulate cortical activity during sleep [21–23], but human evidence is lacking.

These findings position the stomach within a broader interoceptive regulatory loop in which continuous brain-body feedback maintains physiological stability and influences subjective experience [24–26]. While mostly studied in wakefulness, this loop also operates during sleep, coordinating ascending interoceptive signals and descending autonomic control on infraslow timescales [25, 27]. Understanding how this loop engages during NREM sleep could reveal mechanisms by which interoceptive inputs shape arousal and perceived sleep quality. Although polysomnography (PSG) provides an objective measure of sleep architecture, the question of what determines subjective sleep quality remains difficult to answer [28–30], as most perceived restfulness measures overlook the complex brain-body interactions that shape sleep experience. Maladaptive engagement of the interoceptive loop, specifically when visceral or autonomic activations coincide with arousal, may disrupt sleep stability, providing a physiological basis for variability in perceived restfulness. Thus, probing visceral oscillators alongside brain activity can identify new biomarkers of interoceptive integration with cortical and autonomic dynamics.

To address these gaps, we combined high-density EEG and EGG during overnight human sleep to characterize gastric activity and its coupling with cortical dynamics. For the first time, we show that the gastric rhythm itself exhibits infraslow amplitude modulations, revealing hierarchical organization of visceral activity. Furthermore, we demonstrate that stomach-brain coupling is strongest during thalamocortical markers of arousal, and its magnitude relates to how rested individuals feel after sleep. Together, these findings identify the stomach as a key peripheral oscillator in the interoceptive regulatory loop and uncover a previously unrecognized pathway linking visceral physiology, arousal regulation, and human sleep experience.

2 Results

Simultaneous scalp EEG and gastric slow wave activity, collected via EGG, were analyzed in 60 healthy participants (44 female; average age of 23.1 $\pm$ 2.8 years; average BMI 22.5 $\pm$ 3.9 kg/m^2) across three nights (Fig. 1a). The nights consisted of an adaptation night and two experimental sessions, during which controlled awakenings

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were performed to collect dream data; additionally, one of these sessions also included targeted memory reactivation (TMR) [31, 32]. We analyze three complementary gastric markers: 1) average gastric-band power (0.03 to 0.07 Hz [14]) across sleep cycles, 2) infraslow oscillations (ISO) of the gastric power envelope, capturing minute-scale modulation, and 3) changes in gastric amplitude aligned to discrete cortical events, capturing second-scale dynamics.

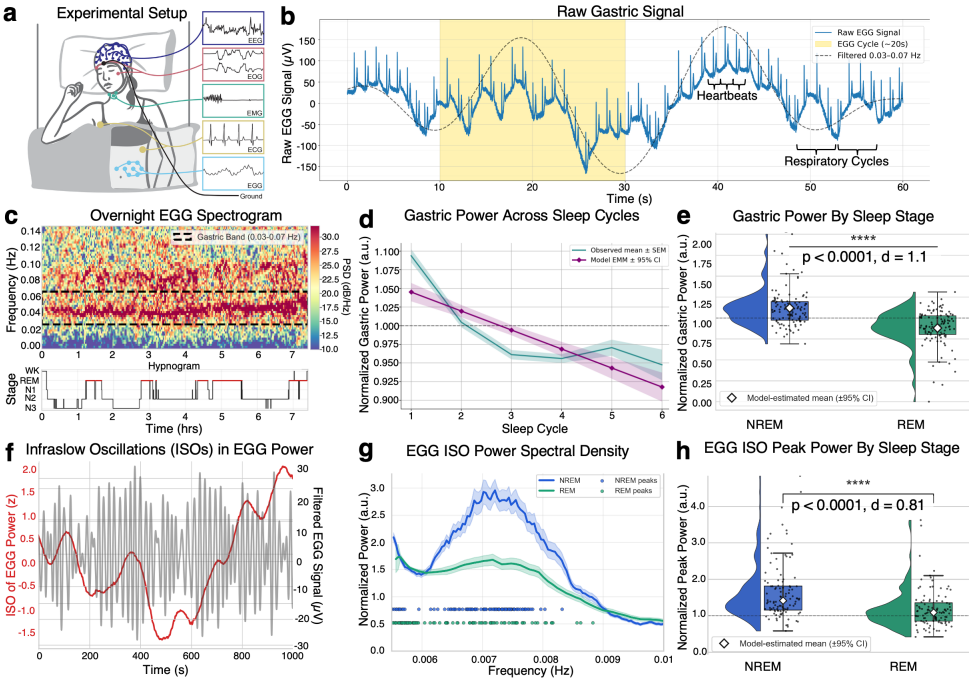


Fig. 1 **(a)** Simultaneous non-invasive electrophysiological recordings included 64-channel EEG, 2-channel EOG, 3-channel EMG, 2-channel ECG, and 4-channel EGG. **(b)** Example of a raw gastric slow wave signal recorded from one participant. The ∼20 second EGG cycle is visible in the time series (yellow shaded region; dashed line represents the band-pass filtered 0.03-0.07 Hz signal). **(c)** Overnight EGG spectrogram in one participant showing sustained power in the normogastric frequency range (0.03 to 0.07 Hz, dashed box). The corresponding hypnogram is shown below. **(d)** Visualization of gastric slow wave power declining across sleep cycles (n = 95 nights). Observed mean ± SEM (standard error of the mean; teal) and model-estimated marginal means (EMMs) ± 95 % CI (magenta; linear mixed-effects model (LMM) controlling for sex, condition, and time-since-meal; $p < 0.0001$). **(e)** Across 95 nights, night-mean normalized gastric-band power was significantly higher during NREM than REM sleep (LMM: $p_{LMM} < 0.0001$ for stage effect, $d_{LMM} = 1.1$). Diamonds and error bars indicate model-EMMs ± 95% CI. **(f)** Example of filtered gastric slow-wave signal (gray) and corresponding infraslow oscillations (ISOs) in EGG power (red; z-scored), showing slow rhythmic fluctuations in gastric amplitude over time. **(g)** Group-level normalized EGG ISO power spectra across sleep stages (n = 104 nights). A clear infraslow peak (∼0.007 Hz; period of ∼2.5 minutes) was more pronounced during NREM sleep. Shaded regions denote ± SEM. **(h)** Normalized EGG ISO peak power across nights with model-EMMs ± 95 % CI that indicate significantly higher ISO peak power during NREM than REM sleep (n = 104 nights, $p_{LMM} < 0.0001$, $d_{LMM} = 0.81$).

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2.1 Gastric electrophysiology tracks sleep architecture: sustained power and infraslow modulation are amplified in NREM stages

Healthy human sleep is organized into recurring sleep cycles, typically four to six per night, each lasting approximately 90 to 110 minutes and comprising successive NREM and REM stages [33]. To characterize how gastric activity varies across sleep stages, we examined stage- and cycle-dependent changes in gastric slow wave power across the night.

Accounting for repeated measures within individuals, sex, experimental condition, and time since last meal, gastric slow-wave power was highest during the first sleep cycle and progressively declined across successive cycles (linear mixed-effects model (LMM): $\beta_{\text{cycle}} = -0.027 \pm 0.003$, $p_{LMM} < 0.0001$; Fig. 1d, Table 1, Fig. 6a-d). This decline was consistent across participants, reflecting a gradual attenuation of gastric activity over the course of the night. The stage $\times$ cycle interaction was not significant ($\chi^2 = 5.89$, $p = 0.21$; Fig. 6c), indicating that the reduction in gastric power across cycles occurred independently of sleep stage.

Time since the last meal also showed a modest but significant negative association with gastric power ($\beta_{\text{mealTime}} = -0.003 \pm 0.001$, $p_{LMM} = 0.006$), suggesting that gastric emptying of the last meal consumed before sleep may contribute to the gradual overnight decline. Neither sex nor experimental condition significantly predicted gastric power (all $p > 0.3$).

Across all nights, gastric power differed between NREM and REM sleep, with REM sleep showing lower power on average ($\Delta_{\text{NREM–REM}} = 0.240 \pm 0.043$, $z = 5.56$, $p_{LMM} < 0.0001$, $d_{LMM} = 1.1$; Fig. 1e, Table 2). However, when stage was recoded as a binary NREM-REM contrast in a separate model (Table 3), several stage $\times$ cycle interaction terms were significant (cycles 2, 4, and 5; all $p < 0.02$), indicating that the magnitude of the NREM-REM difference varied modestly across cycles.

Interpretively, this pattern is descriptive: the decline across cycles may reflect homeostatic processes, progressive gastric emptying, circadian influences, or a combination of these factors, while the NREM-REM difference likely reflects stable autonomic differences between sleep stages.

Given this overnight modulation of gastric power, we next asked whether the gastric signal also exhibits rhythmic fluctuations at infraslow timescales, analogous to the well-characterized infraslow oscillations (ISOs; < 0.1 Hz) previously described in cortical sigma power [6, 8], heart rate [6, 8], respiration [9], and even peripheral glucose [10]. Such infraslow dynamics in the brain and body reflect alternating periods of high and low arousability and have not been examined in the human stomach.

Unlike heart rate, glucose, or respiration, which are scalar physiological outputs, the gastric slow wave is itself a continuous electrophysiological oscillation generated by the interstitial cells of Cajal. This makes it more analogous to an EEG signal, in which the amplitude, or power, of a fast oscillation can fluctuate over slower timescales. Examining infraslow modulations in gastric power therefore tests whether the stomach's intrinsic electrical rhythm is organized hierarchically over infraslow timescales,

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similar to how cortical sigma power waxes and wanes during NREM sleep [8]. Detecting such ISOs in the EGG would suggest that intrinsic gastric rhythms participate in the same slow regulatory framework that coordinates brain and body states.

To evaluate this, we confirmed the presence of ISOs in the gastric signal across sleep stages for the first time (Fig. 1f). The envelope of the filtered gastric slow wave (0.03 to 0.07 Hz) revealed prominent infraslow amplitude modulations at ∼0.007 Hz (∼2.5 minute cycles) during both NREM and REM sleep (Fig. 1g). Group-level power spectra showed a distinct infraslow peak (Fig. 1g). A paired LMM, accounting for repeated measures within nights, sex, condition, and time since last meal, confirmed that normalized EGG ISO power was significantly greater in NREM relative to REM ($\beta = 0.48 \pm 0.08$, $p_{LMM} < 0.0001$, $d_{LMM} = 0.81$; n = 104 nights; Fig. 1g, h). Model-estimated marginal means (±95% CI) further demonstrated consistently higher peak ISO power during NREM, indicating that infraslow gastric rhythmicity is selectively amplified during NREM sleep.

Comparable infraslow structure was also detected during quiet wakefulness, which we defined as continuous, artifact-free wake epochs surrounding sleep onset and offset and, when available, sufficiently long wake bouts during the night to permit ISO calculation. Although ISO structure was present in wake, its power was strongest in NREM sleep (LMM: $\beta = 0.23 \pm 0.10$, $p_{LMM} = 0.01$, $d_{LMM} = 0.33$; Fig. 7a, b). Model-estimated marginal means (±95% CI) further demonstrated that infraslow gastric power rhythmicity was selectively amplified during NREM sleep, reflecting a stronger and more temporally stable coupling of gastric power fluctuations to underlying sleep architecture.

Together, gastric activity during healthy sleep shows: 1) a gradual decline across cycles, beyond just the effect of increasing time since last meal, 2) an overall increase in gastric power during NREM sleep, with modest cycle-dependent variation, and 3) continuous gastric ISOs whose amplitude is enhanced during NREM.

2.2 Infraslow stomach-brain coupling is strongest during N3 sleep

Having established the presence of EGG ISOs, we next asked whether these slow gastric fluctuations co-vary with cortical infraslow activity. Specifically, we compared the EGG ISO with infraslow fluctuations in EEG sigma power (9 to 16 Hz) [34]—a cortical marker of spindle and arousal-related dynamics [7, 8].

While sigma activity peaks during N2 sleep, it remains reliably present during N3 (Fig. 7c; Table 4), where spindles co-occur with slow wave oscillations (SWOs). Furthermore, robust stage-dependent differences in infraslow modulation of sigma activity across NREM stages were observed. Consistent with prior work showing reduced but persistent sigma dynamics in deep sleep [8, 35], sigma ISO power was significantly reduced in N3 relative to N2 ($\beta = -0.67 \pm 0.17$, $p < 0.001$; Fig. 7d; Table 5), while N1 did not differ significantly from N2 ($\beta = 0.13 \pm 0.21$, $p = 0.55$; Fig. 7d; Table 5). Importantly, these effects remained unchanged when controlling for the number of available samples per stage ($p = 0.71$; Table 5), indicating that the reduced sigma ISO modulation in N3 cannot be attributed to differences in stage duration or sampling density. Together, these analyses confirm that both sigma power and sigma-band ISOs

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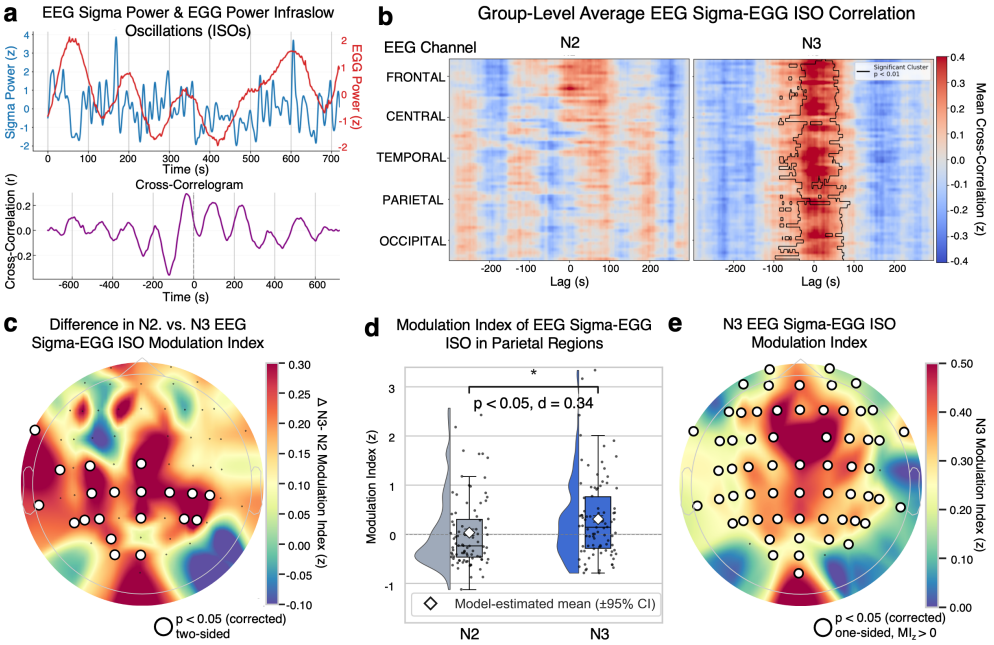


Fig. 2 **(a)** Example trace showing coordinated fluctuations between EEG sigma power (blue; z-scored) and EGG gastric power (red; z-scored) during NREM sleep within the infraslow range, with corresponding cross-correlogram. **(b)** Group-level EEG-EGG cross-correlation matrices across EEG channels during N2 (left) and N3 (right) sleep (n = 106 nights). Significant clusters ($p < 0.01$, cluster-corrected) indicate sustained EEG sigma-EGG ISO coupling during N3 sleep, centered at zero lag for visualization. **(c)** Topographical distribution of differences in modulation index (MI) during N2 and N3, with significantly increased coupling localized to parietal regions (highlighted white regions: $p < 0.05$, two-sided, corrected; n = 106 nights). **(d)** LMM restricted to parietal regions revealed stronger coupling during N3 compared with N2 sleep ($p_{LMM} = 0.033$, $d_{LMM} = 0.34$; n = 91 nights). **(e)** Topographical distribution of EGG ISO phase and EEG sigma amplitude MI during N3 sleep, revealing significant spatial clusters of coupling (highlighted white regions: $p < 0.05$, one-sided, corrected; n = 106 nights).

are present and statistically well supported in both N2 and N3, providing a justified cortical signal for subsequent coupling analyses.

We next compared gastric-cortical infraslow coupling across N2 and N3 sleep to test for stage-dependent differences in gut-brain coordination. During representative bouts of NREM sleep, EEG sigma-band ISO closely tracked the EGG ISO, yielding a clear oscillatory structure in their cross-correlogram (Fig. 2a). Extending this to the group level, cross-correlograms were computed between sigma bandpower in each EEG channel and EGG power, averaged across 5-minute bouts of N2 and N3 sleep. For group statistics and visualization, we corrected the lag axis so that 0 seconds reflects simultaneous window centers. Group-level maps revealed globally significant clusters ($p < 0.01$, corrected) only during N3 (Fig. 2b), indicating temporal coupling between cortical EEG sigma ISO activity and EGG ISO during N3 sleep. In contrast, no significant temporal coordination was observed during N2 (Fig. 2b), N1, REM, or

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wakefulness (Fig. 8a, b). The mean z-scored maximal correlations during N2 and N3 are presented in Table 6.

We then tested whether this infraslow gastric-cortical correlation reflected phase-amplitude coupling (PAC) between the phase of the EGG ISO and the amplitude of EEG sigma ISOs using the modulation index (MI) (Fig. 2c) [36]. A two-sided cluster-based permutation test ($p < 0.05$, corrected) revealed spatially significant differences between N2 and N3, with stronger MIs localized over parietal electrodes, indicating enhanced gastric-sigma coupling during N3 sleep (Fig. 2c). Although absolute MI values are numerically small by definition, even modest non-zero values reflect robust PAC relative to surrogate-based null distributions, which are typically centered near zero [36, 37]. To confirm this finding while accounting for individual and contextual covariates, LMM restricted to the parietal ROI showed a significant increase in MI during N3 relative to N2 ($\Delta_{\text{N3–N2}} = 0.232 \pm 0.108$, $z = 2.14$, two-sided $p = 0.033$, $d_{LMM} = 0.34$; Fig. 2d; Table 7).

ROI-wise models (Table 8, Fig. 9b) confirmed that these stage effects were strongest in parietal ($\Delta = 0.234$, $p = 0.028$, $d_{LMM} = 0.35$) and temporal electrodes ($\Delta = 0.231$, $p = 0.030$, $d_{LMM} = 0.35$), with a trend in occipital electrodes ($\Delta = 0.196$, $p = 0.116$, $d_{LMM} = 0.25$).

Finally, a one-sided cluster-based permutation test ($p < 0.05$, corrected) within N3 identified widespread positive MI clusters ($MI_z > 0$) spanning frontal, central, and parietal areas (Fig. 2e), demonstrating globally elevated gastric-sigma coupling during N3 sleep.

All together, these results show that gastric and cortical sigma ISO rhythms are temporally coordinated, with the most sustained synchrony during N3 sleep. Furthermore, we also indicate that gastric-cortical infraslow PAC is strongest and most spatially widespread during N3, with posterior regions exhibiting the largest stage-dependent increases.

2.3 Cortical slow wave oscillation-spindle complexes drive larger, phasic increases in gastric amplitude

Since gastric and cortical signals exhibit coherent ISOs that are strongest during N3 sleep, we next asked whether transient cortical events that define NREM architecture, slow wave oscillations (SWOs; 0.5 to 4 Hz) and sleep spindles (9 to 16 Hz) [34, 38, 39], also modulate gastric activity. Whereas the previous analyses focused on continuous infraslow coupling between the EGG and EEG (captured through ISO phase-amplitude interactions), here we examined short-lived, event-locked fluctuations in gastric amplitude aligned to the timing of discrete cortical events. This approach allowed us to determine whether the slow, rhythmic coordination observed at the infraslow timescale is mirrored by faster, phasic modulations of gastric activity around uncoupled SWOs and coupled SWO-spindle complexes.

Using event-triggered analyses, we quantified changes in absolute EGG amplitude aligned to the negative peak of cortical SWOs (Fig. 3a). LMMs revealed a robust post-trough increase in EGG amplitude across all NREM events ($\Delta_{\text{Post}-\text{Pre}} = 0.26 \pm 0.01$, $z = 46.8$, $p < 0.001$; n = 91 nights; Fig. 3b, Table 9), indicating a consistent rise in gastric activity following SWO troughs.

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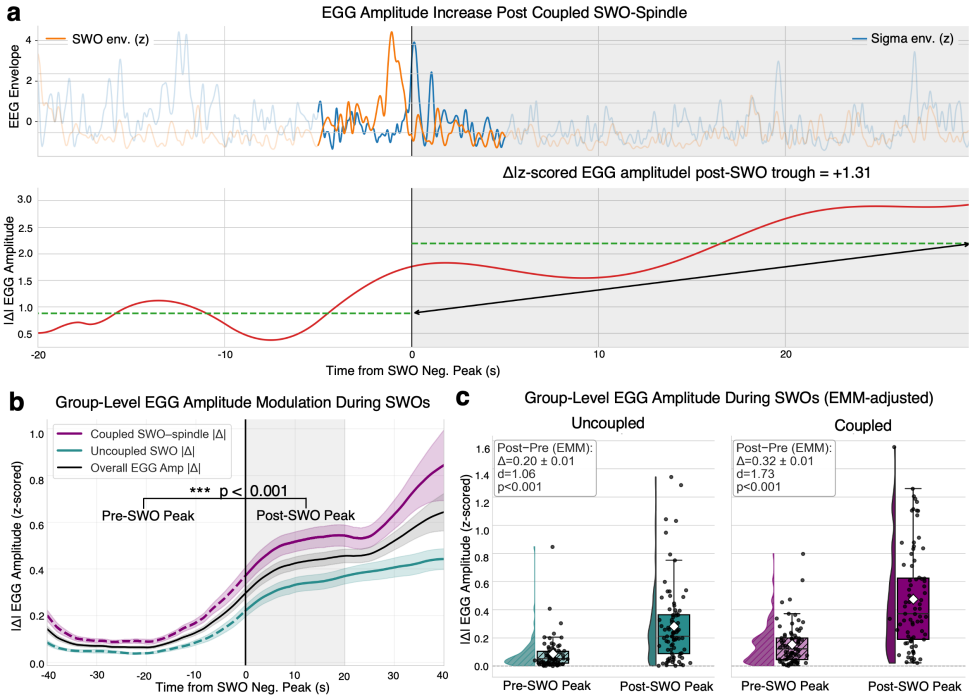


Fig. 3 **(a)** Data from one participant demonstrating coupled slow wave oscillation (SWO)-spindle event and changes in EGG amplitude. The top panel presents z-scored envelopes of the SWO oscillation, followed by the coupled sleep spindle. The bottom panel illustrates the sharp increase in EGG amplitude following the SWO trough (shaded post window indicates 0 to 30 seconds). **(b)** Group-level time course of absolute EGG amplitude modulation (z-scored) aligned to the negative peak of SWOs. Both coupled and uncoupled events exhibited stronger EGG amplitude modulation after the SWO negative peak, with a significant post-trough increase (LMM: $p_{LMM} < 0.001$; $n = 91$ nights). Shaded regions denote ± SEM. **(c)** EMM-adjusted pre-post SWO trough comparisons showing enhanced gastric amplitude modulation following SWO troughs (LMMs, uncoupled: $p_{LMM} < 0.001$, $d_{LMM} = 1.06$; coupled: $p_{LMM} < 0.001$, $d_{LMM} = 1.73$; n = 91 nights). A significant window × coupling interaction ($p_{LMM} < 0.001$) confirmed that coupling strengthened post-SWO gastric modulation. Shaded regions denote ± SEM.

Critically, the model identified a significant window × coupling interaction ($\beta = -0.031 \pm 0.003$, $z = 11.3$, $p_{LMM} < 0.001$; $n = 91$ nights; Fig.3c, Table 9), demonstrating that this post-trough increase was stronger for SWOs that were temporally coupled with sleep spindles. A complementary model of post-pre differences confirmed that EGG amplitude changes were significantly greater for coupled than with uncoupled events ($\Delta_{\text{Coupled–Uncoup}} = 0.125 \pm 0.009$, $p_{LMM} < 0.001$, $d_{LMM} = 0.61$, Table 10).

Model-estimated marginal means (± 95% CI) further showed significant post-pre increases for both uncoupled (EMM: $\Delta_{\text{Post}-\text{Pre}} = 0.20 \pm 0.01$, $p_{LMM} < 0.001$, $d_{LMM} = 1.06$; Fig. 3c, Table 10) and coupled ($\Delta_{\text{Post}-\text{Pre}} = 0.32 \pm 0.01$, $p_{LMM} < 0.001$, $d_{LMM} = 1.73$; Fig. 3c, Table 10) SWO events, confirming that coupling markedly amplified the magnitude of gastric modulation. None of the covariates, namely, experimental condition, sex, or time since last meal, were significant in any model (Table 9, 10).

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Collectively, these results indicate that gastric amplitude is dynamically modulated in relation to cortical SWOs, with the strongest enhancement following SWO troughs that co-occur with spindles. These brief, event-locked increases provide a plausible building block for the sustained ISO coordination observed above.

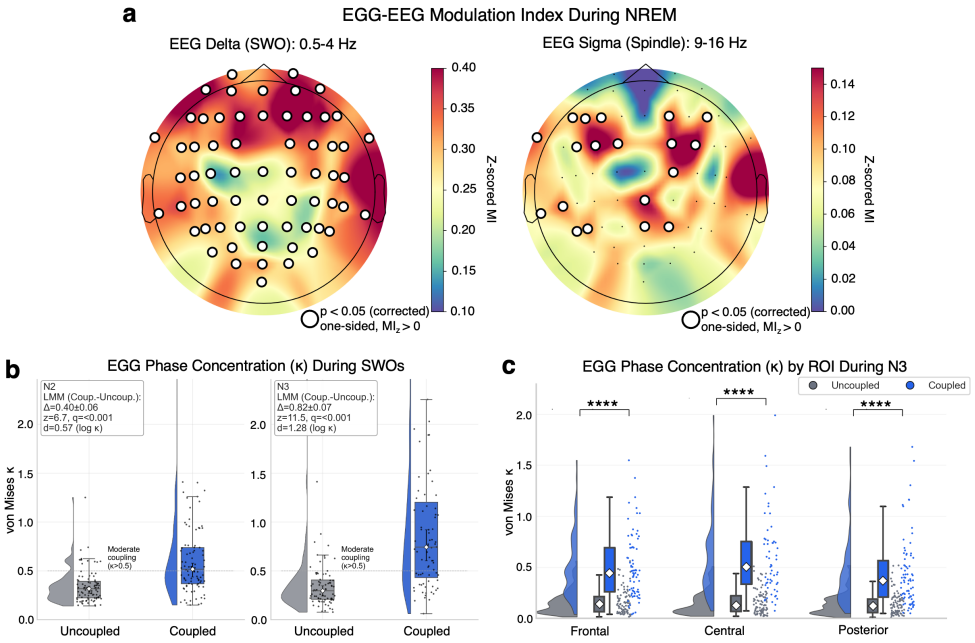


Fig. 4 **(a)** Topographical maps of PAC between EGG phase and EEG amplitude, revealing significant clusters in delta (0.5-4 Hz) and sigma (9-16 Hz) bands during NREM (highlighted white regions: $p < 0.05$, corrected; n = 106 nights). **(b)** Group-level distributions of gastric phase concentration (κ) during coupled and uncoupled SWO events across N2 and N3 sleep (n = 91 nights). LMMs confirmed a robust main effect of coupling (left, N2: $p < 0.001$, corrected; right, N3: $p < 0.001$, corrected; n = 91 nights), corresponding to approximately 2-fold and 4-fold increases in gastric κ, respectively. **(c)** Regional analysis of N3 SWOs showing strong and spatially uniform coupling effects across the frontal, central, and posterior electrodes (all $p < 10^{-6}$, corrected; 3.4 to 4.8 fold increases in gastric κ; n = 91 nights).

2.4 Gastric phase coupling during NREM slow wave oscillation and spindle events

2.4.1 Gastric phase is coupled to cortical delta and sigma frequencies during NREM sleep

We asked whether amplitude modulations reflect phase-specific coordination between gastric and cortical activity. As such, we assessed cross-frequency PAC between the EGG phase and EEG amplitude across canonical frequency bands [36].

Significant MIs were observed in the delta (0.5 to 4 Hz [40]) and sigma (9 to 16 Hz [34]) bands during NREM sleep ($p < 0.05$, corrected; $n = 106$ nights; Fig. 4a),

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indicating that the amplitude of slow wave and spindle EEG activity is systematically coupled with the phase of the gastric slow wave.

2.4.2 Gastric phase alignment is strongest during N3 slow wave oscillations

To quantify the strength and consistency of stomach-brain coordination, we computed the von Mises concentration parameter, κ, [41] as a measure of EGG phase alignment during cortical SWOs. Across all nights, gastric phase concentration was significantly higher during coupled SWO-spindles compared to uncoupled SWOs (Fig. 4b). This enhancement was especially pronounced in N3 sleep (LMM: $\beta = 0.82 \pm 0.07$, $z = 11.5$, corrected $p_{LMM} < 0.001$, $d_{LMM} = 1.28$; n = 91 nights; Fig. 4c, Table 11), corresponding to an approximately four-fold increase in EGG κ. In contrast, the coupling effect during N2 was smaller but still significant (LMM: $\beta = 0.40 \pm 0.06$, $z = 6.7$, corrected $p_{LMM} < 0.001$, $d_{LMM} = 0.57$; Fig. 4b, Table 12). Across sleep cycles, EGG κ values remained stable in N2 but increased progressively during N3 (LMM: $\beta = 0.11 \pm 0.04$, $p_{LMM} = 0.018$; Fig. 10a; Table 13), suggesting a gradual strengthening of stomach-brain phase alignment as sleep deepened.

2.4.3 Gastric phase alignment during coupled slow wave oscillations is globally distributed across the human cortex

We next examined the spatial pattern of this coupling effect across cortical regions. LMMs revealed a robust main effect of coupling (all $p < 10^{-6}$, corrected), with the magnitude of the effect comparable across the frontal ($\beta = 1.22$, 3.38 fold increase), central ($\beta = 1.56$, 4.78 fold increase), and posterior ($\beta = 1.32$, 3.73 fold increase) regions (n = 91 nights; Fig. 4c; Table 13). These findings indicate that gastric phase locking during N3 coupled SWOs is not localized but rather global, suggesting synchronous modulation of cortical slow wave networks and peripheral gastric rhythms.

Consistent with this, regional analyses revealed that the coupling effect persisted across all EEG channels in N2 (all $p < 10^{-6}$ corrected, n = 91 nights; Fig. 10b). Furthermore, within coupled SWOs, EGG κ values increased during N3 compared to N2, specifically in frontal ($\beta = 0.19$, $p = 0.036$ corrected, 1.21 fold increase, n = 91 nights; Fig. 10c) and posterior regions ($\beta = 0.26$, $p = 0.0054$ corrected, 1.29 fold increase, n = 91 nights; Fig. 10c). These results demonstrate that SWO-spindle complexes during N3 represent the state of strongest and most globally distributed gastric phase locking, reflecting a large-scale coordination of cortical slow wave networks and visceral rhythms.

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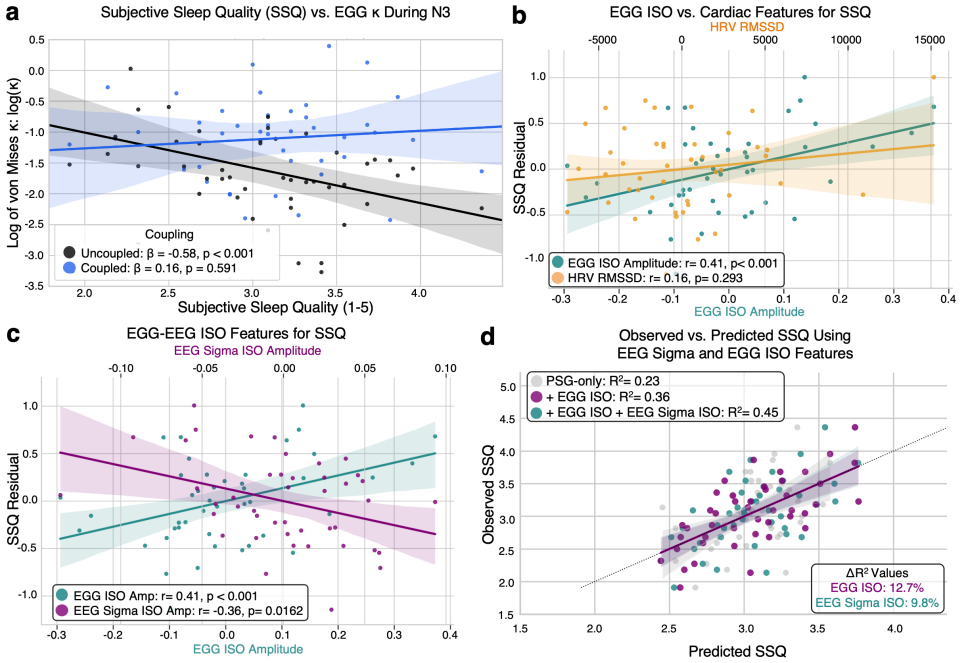


Fig. 5 **(a)** Across participants, greater gastric phase alignment during uncoupled SWOs in N3 was associated with lower subjective sleep quality (SSQ) scores ($p = 0.0007$; n = 49 nights), whereas no significant association was observed for coupled SWOs ($p = 0.59$; n = 49 nights). **(b)** Partial correlations between SSQ, cardiac and EGG ISO predictors after controlling for PSG features. Across nights, variance in the EGG ISO (blue) showed a robust positive association with SSQ ($r = 0.41$, $p = 0.006$; n = 49 nights), whereas heart-rate variability (HRV; RMSSD; orange) showed no reliable relationship ($r = 0.16$, $p = 0.29$; n = 49 nights). **(c)** Partial correlations between SSQ, EGG and cortical sigma ISO predictors. Both variance in the EGG ISO (blue, $r = 0.39$, $p = 0.008$) and EEG sigma ISO (green, $r = -0.37$, $p = 0.016$; n = 49 nights) showed a robust inverse association with SSQ. **(d)** Integrating both EGG and EEG ISO features with polysomnography (PSG) metrics markedly improved model fit by two fold ($\Delta R^2 = 0.45$; $\Delta R^2_{EGG} = 0.13$, $R^2_{EEG\sigma} = 0.10$; n = 49 nights).

2.5 Functional relevance of gastric and stomach-brain dynamics for sleep quality

2.5.1 Stronger gastric phase alignment during uncoupled N3 slow wave oscillations predicts poorer sleep quality

Having established that gastric rhythms exhibit strong and spatially widespread phase alignment during N3 SWOs (Fig. 4c), we next examined whether this coordination was functionally related to measures of sleep quality. Focusing on the adaptation night, where sleep was least affected by experimental manipulation, we quantified associations between gastric phase concentration (κ) during cortical SWO events and both subjective and objective measures of sleep quality.

Across participants, greater gastric phase alignment during uncoupled SWOs in N3 was significantly associated with lower subjective sleep quality (SSQ) scores (LMM: $\beta = -0.58 \pm 0.14$, $p = 0.0007$, n = 49 nights; Fig. 5a), whereas no significant association

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was observed for coupled SWOs (LMM: $\beta = 0.16 \pm 0.29$, $p = 0.59$, n = 49 nights; Fig. 5a).

To assess whether this effect extended to objective indices of sleep architecture, we repeated the analysis across both N2 and N3 sleep stages for subjective ratings, sleep maintenance efficiency (SME), total sleep time (TST), and wake after sleep onset (WASO) (Fig. 11b; Table 14). Consistent with the subjective findings, only N3 uncoupled events showed a significant negative association objective measures. Stronger N3 uncoupled κ predicted worse SME (LMM: $\beta = -0.18$, corrected $q = 0.009$, n = 49 nights; Fig. 11b; Table 14), shorter TST (LMM: $\beta = -27.36$, corrected $q < 0.05$, n = 49 nights; Fig. 11b; Table 14) and greater WASO ($\beta = 28.06$, corrected $q < 0.05$, n = 49 nights; Fig. 11b; Table 14), whereas coupled events showed no reliable relationships.

Together, these results demonstrate that the strength and context of stomach-brain coupling may carry functional relevance for sleep quality. Specifically, increased gastric phase locking during uncoupled slow wave events is linked to poorer subjective and objective sleep quality, while coupling with spindles, reflecting more coordinated cortical processing, seemed to have no association.

2.5.2 EGG ISO variance uniquely explains subjective sleep quality beyond polysomnographic and cardiac measures

To test whether the strength of gastric-cortical coupling explains unique variance in SSQ beyond standard polysomnographic (PSG) metrics, we compared a series of nested linear models.

A baseline model including macro-architectural NREM sleep features (%N1, %N2, %N3, %Wake), sleep-fragmentation indices (sleep to wake and N3 to N1/N2 transitions per hour), and other covariates, such as sex, experimental condition, time since last meal, and coupling condition, explained 22.5% of the variance in SSQ ($R^2 = 0.225$, adjusted $R^2 = 0.133$; $p = 0.017$). Adding EGG κ during N3 uncoupled SWO events did not improve model fit ($\Delta R^2 = 0.0005$, partial $R^2 = 0.0005$, $p = 0.84$) and left all PSG coefficients virtually unchanged (Table 15).

Cross-validated performance yielded similarly negligible differences ($\Delta \mathrm{CV} R^2 = -0.015$). Stepwise model comparisons confirmed that while adding sleep-fragmentation indices modestly increased explained variance ($\Delta R^2 = +0.045$, $p = 0.14$), the inclusion of EGG κ or its interaction with coupling status had no measurable effect ($\Delta R^2 \leq 0.018$, all $p > 0.20$).

Next, we examined whether intrinsic features of the gastric slow wave alone, reflecting autonomic tone independent of cortical coupling, could explain additional interindividual variance in SSQ. For each night, we quantified a set of autonomic EGG metrics encompassing total spectral power (0.01 to 0.15 Hz), the proportion of brady-, normo-, and tachygastric states, and the amplitude variance of the EGG ISO activity presented prior.

Across participants, the infraslow EGG amplitude variance showed the strongest positive association with SSQ (partial $r = 0.41$, $p = 0.006$, n = 49 nights; Fig. 5b; Table 16), surpassing all other gastric and cardiac metrics. Adding this single EGG measure to the baseline PSG model significantly improved explained variance, accounting for

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an additional 13% of the variance in SSQ ($\Delta R^2 = 0.13$, partial $R^2 = 0.16$, $p = 0.014$; Table 16; Fig. 5b), whereas neither heart-rate variability (HRV; RMSSD) nor heart-rate infraslow power produced meaningful gains ($\Delta R^2 < 0.03$, all $p > 0.25$; Table 16).

A direct model comparison confirmed that EGG features accounted for a greater proportion of variance in perceived sleep quality than canonical cardiac autonomic indices. Including HRV alongside EGG ISO amplitude variance did not further enhance predictive power ($R^2 = 0.37$ vs. $R^2 = 0.36$; Table 16). In contrast, the phase-based EGG κ measure, indexing stomach–brain synchrony during NREM sleep, explained negligible additional variance ($\Delta R^2 < 0.001$, $p = 0.84$; Table 16), consistent with its overlap with standard PSG measures of sleep continuity.

Thus, intrinsic gastric power ISO dynamics, not its coupling with the cortex or cardiac variability, provide a unique and powerful autonomic signature of perceived sleep quality, capturing variance in subjective restoration beyond that explained by traditional PSG and HRV metrics.

2.5.3 Complementary infraslow dynamics in the stomach and cortex jointly predict subjective sleep quality

To determine whether infraslow cortical dynamics contribute uniquely to perceived sleep quality, we compared models incorporating gastric and EEG infraslow features. The infraslow amplitude variance of cortical sigma power (<0.1 Hz) showed a significant negative association with SSQ (partial $r = -0.37$, $p = 0.016$, $n = 49$ nights; Fig. 5c; Table 17), comparable in magnitude but opposite in direction to the positive relationship observed for gastric ISO amplitude variance.

Adding either feature to the baseline PSG model significantly improved explained variance in SSQ (EGG ISO: $\Delta R^2 = 0.13$, $p = 0.019$; EEG sigma ISO: $\Delta R^2 = 0.11$, $p = 0.031$; Fig. 5d; Table 17), whereas coupling strength between the two rhythms, quantified by the parietal modulation index (MI), did not yield a significant improvement ($\Delta R^2 = 0.046$, $p = 0.17$; Table 17).

Notably, combining both EGG and EEG ISO features more than doubled the variance explained relative to PSG alone ($\Delta R^2 = 0.23$, $p = 0.0047$; n = 49 nights; Fig. 5d; Table 17), indicating that variations in gastric and cortical ISOs contribute complementary and nonredundant information about perceived sleep quality. These results suggest that consistency of infraslow modulation within both the brain and stomach, rather than direct stomach-brain coupling, best accounts for interindividual differences in subjective sleep quality.

3 Discussion

The present study demonstrates that the human stomach exhibits structured, stage-dependent rhythmicity during sleep, expressed through sustained power, infraslow modulation, and transient, event-locked changes. Across these complementary measures, gastric activity aligns with sleep architecture and cortical excitability cycles, identifying the stomach as a peripheral oscillator participating in the infraslow regulation of sleep physiology. These findings extend the framework of the interoceptive

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regulatory loop [25], previously characterized in wakefulness, to the sleeping brain, positioning gastric rhythms as a physiologically meaningful component of sleep architecture. By combining high-density EEG with EGG, the present study provides direct electrophysiologic evidence that gastric activity is temporally coupled with cortical dynamics, revealing a hierarchical organization of visceral rhythms across the night.

3.1 Gastric rhythmicity mirrors sleep architecture and homeostatic depth

Gastric slow-wave power varied systematically with sleep architecture, showing its highest values during NREM sleep and a progressive decline across successive sleep cycles. This pattern aligns with prior work indicating that gastric slow-wave activity is regulated by parasympathetic (vagal) efferent input and central autonomic circuits [11]. Increased gastric slow-wave power during NREM sleep is therefore consistent with the well-established shift toward vagal predominance during this stage, reflected in decreased heart rate, reduced sympathetic tone, and increased heart rate variability [42]. At the same time, the overall decline across cycles likely reflects multiple influences, including homeostatic changes, progressive gastric emptying, and circadian factors, suggesting that these stage- and cycle-related patterns should be interpreted as descriptive rather than mechanistic.

The gradual decline in gastric power across the night parallels the decay of homeostatic sleep pressure and the reduction in metabolic demand. While previous studies have reported that gastric motility diminishes overnight as postprandial activity wanes [43], our results indicate that this decline cannot be fully explained by the time since the last meal. The persistence of elevated gastric power throughout the night and its association with sleep stage suggest that these dynamics may reflect, in part, centrally coordinated autonomic regulation rather than solely passive digestive processes. This stage-dependent modulation positions the stomach as a sensitive autonomic indicator of sleep depth and continuity, akin to cardiac and respiratory rhythms but slower and potentially more stable.

3.2 Infraslow organization of gastric activity and its amplification during NREM

A key discovery of this study is that the gastric signal itself exhibits robust infraslow modulations at $\sim$0.007 Hz, on the same timescale as cortical and autonomic ISOs that gate arousability and synchronize brainstem and thalamocortical activity [6–8, 35]. These fluctuations were present across all vigilance states but were selectively amplified during NREM sleep.

Because infraslow rhythms in the central nervous system are partly driven by locus coeruleus noradrenergic cycling and parasympathetic activity [6], the parallel timing observed in gastric activity likely reflects shared regulation through brainstem circuits. The locus coeruleus and dorsal vagal complex both project to gastric smooth muscle [14] and interstitial cells of Cajal and have been shown to modulate motility via noradrenergic inhibition [44, 45]. Thus, the presence of infraslow gastric

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oscillations suggests that the visceral domain participates in the same infraslow regulatory rhythm that structures cortical excitability and autonomic tone during NREM. Another potential explanation for this infraslow stomach-brain coupling may involve shared hemodynamic fluctuations, as quasi-periodic patterns in blood flow and oxygenation have been shown to influence both neural and autonomic activity during low-frequency arousal cycles [46, 47]. Similarly, recent studies using simultaneous EEG and fMRI have demonstrated that coordinated infraslow hemodynamic oscillations, particularly during NREM sleep, closely track neural activity and autonomic state transitions, establishing a direct link between arousal, vascular dynamics, and slow oscillatory patterns across the brain and body [48–50]

3.3 N3 sleep provides the most stable window for stomach-brain coordination

We next found that infraslow coordination between gastric ISOs and EEG sigma-band power was sustained and strongest during N3 sleep. This coupling, further quantified by phase-amplitude interactions between the EGG ISO phase and sigma amplitude, was most pronounced in parietal regions, consistent with characterization of the EEG sigma ISO [8] and known cortical hubs of interoceptive-exteroceptive integration [51].

While sigma power and spindles are typically densest during N2 [52, 53], their stability and synchrony with other systems maximize in N3 sleep. N2 has been linked to local sensory gating and hippocampal-cortical memory processing, whereas N3 represents a globally coherent and parasympathetically dominated state associated with the lowest arousability and maximal restorative function [40, 54]. Prior studies of cortical-interoceptive coherence [19, 55] similarly show that N3 sleep enhances cross-system stability despite lower cortical amplitude of sigma and spindle activity. Our findings align with this view: during stage N3, cortical and visceral oscillators appear to enter a state of global synchrony, perhaps reflecting coordinated parasympathetic control through brainstem hubs such as the nucleus tractus solitarius, dorsal motor nucleus, and locus coeruleus [14, 15, 26, 51]. This organization supports the notion that N3 represents a physiological safe harbor, where whole-body systems operate under unified low-arousal dynamics.

3.4 Event-locked cortical dynamics transiently modulate gastric activity

To understand the mechanisms underlying this infraslow coordination, we examined transient cortical events. Gastric amplitude rose immediately after the trough of cortical slow-wave oscillations (SWOs), with the strongest increases occurring when SWOs co-occurred with spindles. In parallel, gastric phase alignment was highest during SWO-spindle complexes, indicating that the timing of cortical up-down transitions is highly aligned with the phase of the gastric slow wave.

These findings suggest that dynamic parasympathetic adjustments are precisely timed to cortical state transitions. During coupled SWO-spindle events, the cortical upstate coincides with thalamocortical bursts [38, 40], perhaps providing an optimal

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temporal window for communication between forebrain circuits and subcortical autonomic centers. Such transient modulations likely accumulate to produce the sustained infraslow synchrony observed across minutes of N3 sleep. Whether the link between autonomic and visceral activity is descending or ascending can be the subject of future work.

Furthermore, recent studies have shown that hippocampal ripple events, which are nested within SWO-spindle complexes [56], can causally modulate peripheral metabolism through hypothalamic circuits [10, 57, 58]. By analogy, the synchronized cortical events that structure memory consolidation may simultaneously entrain visceral rhythms, aligning neural and autonomic rhythms within a shared temporal framework. Thus, the same microarchitectural events that stabilize cognitive processing during NREM sleep may also propagate physiological stability throughout the body, linking cortical excitability cycles to the regulation of internal homeostasis.

3.5 Functional relevance: gastric and stomach-brain dynamics predict sleep quality

Finally, we demonstrate that the timing and context of stomach-brain coupling are functionally relevant for sleep quality. Stronger gastric-cortical phase locking during uncoupled SWOs, isolated cortical transitions lacking spindle-mediated integration, predicted poorer subjective and objective sleep quality. Uncoupled SWOs are known to accompany fragmented sleep, reduced memory consolidation, and higher microarousal probability [59–61]. High gastric phase alignment in this context may reflect mistimed or maladaptive engagement of interoceptive regulatory loops, where cortical slow waves occur without coherent thalamic gating, leading to unstable central-peripheral communication and poorer perceived restfulness [62].

In contrast, coupling with spindles likely reflects adaptive synchronization that stabilizes interoceptive feedback and supports restorative sleep. Beyond coupling, we show that the amplitude variance of gastric ISOs uniquely predicts subjective restfulness beyond standard PSG and cardiac measures. While HRV-based indices capture general parasympathetic tone, their association with perceived sleep quality is weak once sleep architecture is controlled [63, 64]. Cardiac metrics are constrained by the heart's multiple regulatory roles, whereas stomach-derived ISOs appear to provide a more specific and temporally stable marker of restorative autonomic regulation during NREM. Notably, this gastric ISO feature explained more unique variance in subjective sleep quality than either EEG sigma ISOs or cardiac measures when considered alongside standard PSG features, indicating that slow-timescale gastric dynamics capture aspects of perceived restfulness that are not reflected in canonical central or autonomic indices.

3.6 Limitations and Future Directions

These analyses are inherently correlational, precluding inference about directionality. Future work can employ passive causal inference methods on simultaneously recorded waveforms such as those obtained in these findings using methods such as Granger causality [65] or directed information [66]. Additionally, active causal manipulations

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can be performed, such as gastrointestinal stimulation of vagal afferents [67, 68], vagus nerve stimulation [69], or cortical stimulation [70], to directly test the flow of information between the cortex and gastric slow wave in these sleep contexts.

Methodologically, the present results are based on surface abdominal recordings comprising a few electrodes and cannot provide spatially specific gastric activity or extract spatial propagation parameters of the gastric slow wave. Future studies can explore using of the high-resolution EGG [71], source localization [72], or invasive recordings [73], to resolve spatial propagation patterns within the stomach, including regional differences between the antrum and corpus, which vary in pacemaker amplitude, density and vagal innervation.

Looking forward, this framework opens avenues to examine how cortical–gastric coupling supports cognition during sleep. Given that SWO-spindle complexes drive both memory consolidation and the strongest gastric modulation observed here, future work should test whether the magnitude or timing of gastric slow wave modulation predicts overnight memory retention or learning outcomes. Such analyses could reveal whether interoceptive synchronization contributes not only to physiological restoration but also to cognitive processing during sleep.

Finally, the present framework can be extended to clinical populations in which autonomic fragmentation and poor sleep co-occur, such as insomnia, dysautonomia, or irritable bowel syndrome. Disrupted gastric rhythmicity frequently accompanies disturbed sleep in these conditions, providing a physiological window into the integrity of sleep-autonomic regulation. Understanding how gastric-cortical coherence varies in these conditions may help identify novel physiological biomarkers of sleep continuity and recovery.

Together, the findings of this study reveal that sleep continuity and subjective restoration emerge from coordinated timing between cortical and visceral oscillators operating on shared infraslow timescales. The stomach acts not merely as a passive effector of autonomic output but as a stable peripheral oscillator that reflects, and may contribute to, the rhythmic regulation of arousal and restoration. By extending the interoceptive regulatory loop into NREM sleep, this work bridges stomach physiology with neural mechanisms of sleep continuity, establishing a foundation for future research on how internal bodily rhythms shape the human experience of rest and quiescent states.

4 Methods

Data collected from 60 healthy participants (44 female; average age of 23.1 $\pm$ 2.8 years; average BMI 22.5 $\pm$ 3.9 kg/m^2) who were a part of a study on dreaming and memory consolidation were included for the analysis. All participants provided written informed consent to the study protocol, approved by CMO Regio Arnhem-Nijmegen. They came to the laboratory for three different nights (one adaptation night and two experimental nights). All participants in the study were physically and mentally healthy, confirmed via the Beck Depression Inventory [74] and the Beck Anxiety Inventory [75]; participants with a history of sleep disorders, intake of medication that

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influences sleep/wake cycles, and frequent coffee consumption (greater than 4 cups/-day), were excluded. Further details of the inclusion and exclusion criteria and the experimental design procedure can be found in the study's registered report [76].

4.1 Electrophysiological recordings

Electroencephalography (EEG) was recorded throughout the experiment using a BrainVision acquisition system and a 64-channel actiCAP slim electrode cap configured according to the 10-20 montage [77]. Two electrodes were used to measure electroculography (EOG) and electrocardiography (ECG), and three electrodes were used for chin electromyography (EMG). All electrophysiology data were recorded with a 500 Hz sampling frequency, referenced to the vertex. Electrogastrography (EGG) was collected using four channels in a bipolar setup, using the same layout as described in Rebollo et al 2018 [14].

4.2 Experimental overview and task

During the adaptation night, participants were invited to the Donders (EEG) laboratory. During the application of the EEG and other electrodes, participants answered questionnaires regarding their sleep habits and mood. Then they were provided with a sleep opportunity of 8 hours starting approximately at 11 pm. Inclusion criteria accounted for whether their regular bedtime was at maximum two hours before or after the given sleep opportunity [76]. In the morning, all participants were asked to complete a questionnaire about their sleep quality via the Schlaffragebogen-A (SF-AR) questionnaire [78]. The SF-AR is a 22-question survey used to quantitatively and qualitatively assess sleep behavior and experience through subjective ratings of: difficulty initiating sleep, difficulty maintaining sleep, premature awakening, general sleep characterization, and total sleep duration, as well as factors of sleep quality such as: feeling of restfulness after sleep, psychological balance before bedtime, psychological exhaustion before bedtime, and psychosomatic symptoms during sleep [78].

During the experimental sessions participants additionally completed a word-picture association learning task, previously used in prior studies [31], with three learning blocks and two recall blocks separated by a ten minute break. The two experimental sessions were separated by at least 14 days and were counterbalanced between the participants with random assignment. Both the order and the images used in the task were also randomized among all participants, and all participants were blinded to the experimental session. During the control experimental night, participants were awoken a maximum of four times from NREM and four times from REM sleep, at least 15 minutes after the first start of the respective NREM/REM sleep stage to collect dream reports.

4.3 Scoring of human EEG data and preprocessing methods

EEG recordings were first synchronized and inspected in MATLAB, with all experimental triggers verified and corrected. Preprocessing was then performed in Python using the MNE-Python toolbox. Continuous data were segmented from lights-off to lights-on, bandpass-filtered (0.5 to 40 Hz with a zero-phase Butterworth), and trimmed

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to remove periods without valid data. For sleep staging, signals from F3, F4, C3, C4, O1, and O2 were used unless automatically flagged as noisy by Pyprep [79], in which case neighboring electrodes were substituted. Cleaned recordings were exported as EDF files and scored using the deep-learning algorithm U-Sleep [80], incorporating EEG and EOG channels when available.

To validate automatic scoring, ten experimental nights were independently scored by an experienced human rater. If epoch-wise agreement exceeded 70%, remaining datasets were accepted with targeted manual review limited to low-confidence periods (certainty score < 4 for ≥ 3 consecutive minutes) and 1-minute intervals preceding awakenings that diverged from live annotations. If agreement fell below this threshold, U-Sleep parameters were fine-tuned (e.g., epoch length, channel set) until acceptable reliability was achieved. Final hypnograms were derived from 30 second epochs classified as Wake, NREM (N1, N2, N3), or REM, with all low-confidence epochs re-evaluated by a trained scorer.

EEG and EGG data were then subjected to semi-automatic artifact rejection using the High-Density SleepCleaner toolbox [81]. Artifacts were identified by trained raters in 10 second segments and stored as binary masks. Noisy EEG and EGG epochs were excluded; missing EEG data were interpolated via spherical splines, with a minimum two valid neighbors, and segments with contiguous artifacts exceeding three minutes were discarded. Clean EEG was bandpass filtered between 0.5 to 35 Hz.

EGG segments with continuous artifacts longer than three minutes were omitted from further analysis. The remaining EGG signals were notch-filtered at 50 Hz and denoised using a combined Wiener and adaptive thresholding procedure to remove residual motion and electrical artifacts [82]. To determine the optimal EGG channel, the channel with the most prominent peak within the gastric power band (0.03 and 0.07 Hz) was selected and used for subsequent analyses.

4.4 Extraction of infraslow oscillations in EEG and EGG data

To examine infraslow oscillatory (ISO) activity in both brain and stomach signals, we applied parallel multitaper approaches to EEG sigma power and gastric EGG recordings.

4.4.1 EEG sigma power ISO extraction

Short-time multitaper spectrograms within the EEG sigma frequency, 9 to 16 Hz [34], were applied to all EEG channels, using five tapers with four-second windows, a one-second step size, and a time bandwidth product of three. Sigma band power was obtained by summing spectral power across the sigma frequency range at each time point, yielding a sigma power time series sampled at 10 Hz.

4.4.2 EGG gastric power ISO extraction

Multitaper spectrograms within the gastric frequency range, 0.033 to 0.067 Hz [14], were applied to the EGG signal, using three tapers with 400 second windows, a one second step size, and a time bandwidth product of two. Gastric band power was

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obtained by summing spectral power across the gastric frequency range at each time point, yielding a gastric power time series sampled at 1 Hz.

To characterize ISOs in the gastric power time series, a multitaper spectrogram was computed using five tapers with 300 second windows, a one-second step size, and time bandwidth product of three, focusing on the infraslow frequency range between 0.005 to 0.1 Hz.

4.5 Spatiotemporal cross-correlation of EEG sigma power and EGG power ISOs

EEG sigma power time series were low-pass filtered, with a cutoff of 0.04 Hz, and the EGG power time series was upsampled to 10 Hz. As the EGG power was estimated using 400 second multitaper windows and the EEG sigma power using 4 second windows, the EGG time series was shifted by 198 seconds to align the centers of the respective analysis windows. Both signals were segmented into non-overlapping, continuous five-minute bouts across sleep stages. Within each bout, EEG sigma power and EGG power were z-scored, and cross-correlation functions were computed across symmetric time lags. Bout-level cross-correlograms were averaged within each EEG channel, yielding subject-specific spatial cross-correlograms per stage. For group-level comparisons, subject-level cross-correlograms were z-scored across lags and averaged across participants.

Statistical assessment of spatiotemporal clusters in the cross-correlograms were identified using non-parametric spatiotemporal cluster-based one-sample t-test, implemented in MNE-Python. The test incorporated both sensor-space adjacency and temporal adjacency between neighboring lag samples to identify clusters that were contiguous in both space and time. Clusters were formed using a cluster-forming t-threshold corresponding to $p = 0.05$ (two-tailed), and cluster-level significance was evaluated with 5,000 permutations at $\alpha = 0.01$ (two-tailed, cluster-level corrected).

4.6 Event detection in human EEG data

Sleep spindles and slow wave oscillations (SWOs) were identified using the YASA package in Python [83], with outliers removed using the package's isolation forest algorithm for anomaly detection in data. Spindles were detected in all clean EEG channels between 9 to 16 Hz and were selected if its duration lasted between 0.5 to 2 seconds, minimum distance between two spindles is 500 milliseconds, and fit the default relative power, moving correlation and root mean square detection thresholds [83]. SWOs were detected in all clean EEG channels between 0.5 to 4 Hz and were selected if the minimum and maximum 1) duration of the negative deflection fell between 0.3 to 1.5 seconds, 2) duration of the positive deflection fell between 0.1 to 1 second, 3) amplitude of the absolute negative trough fell between 40 to 200μV, 4) amplitude of the absolute positive peak fell between 10 to 150μV, and 5) amplitude of the peak to peak fell between 75 to 350μV [83].

To isolate SWO-spindle complexes, the SWO's trough was marked at the timing of its negative peak. A SWO was marked as coupled if a spindle peak occurred within 1.5 second after the downstate, and uncoupled otherwise [38]. A spindle was marked

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as coupled if it occurred within 1.5 seconds after the SWO negative peak, or during the SWO's up-state. This 1.5 second window encompasses the empirically observed rebound and grouping of spindles during the up-state, which typically extends from several hundred milliseconds to over one second after the SWO negative peak [38]. Event detection was isolated to stages N2 and N3.

4.7 Gastric-sleep oscillation coupling using modulation index

Phase-amplitude coupling (PAC) between gastric and cortical oscillations were calculated using the modulation index (MI) [36]. All analyses were restricted to NREM sleep (N2 and N3), requiring at least five minutes of artifact-free data per night.

4.7.1 EGG power ISO phase and EEG sigma power ISO amplitude modulation index parameters

For EGG ISO-EEG sigma ISO PAC, the EGG and EEG sigma power time series extracted from the multitaper method above was used. Each signal was band-pass filtered with zero-phase Butterworth filters, with the EGG ISO phase derived from 0.005 to 0.009 Hz and the EEG sigma ISO amplitude derived from 0.006 to 0.08 Hz in 0.002 Hz steps. Analytic signals were obtained with the Hilbert transform, and only continuous epochs of more than 5 minutes of either N2 and N3 sleep were included in the analysis.

4.7.2 EGG phase and EEG amplitude modulation index parameters

For EEG-EGG PAC, the EGG signal was narrow-band filtered at the gastric power band (0.03 to 0.07 Hz) using a zero-phase fourth-order Butterworth filter. The instantaneous phase was obtained from the analytic signal via the Hilbert transform. For each EEG channel, narrowband amplitude enveloped (0.5 to 25.5 Hz in 1 Hz steps) were extracted using the same filter parameters and Hilbert transform. Filtering and Hilbert transforms were performed on the full continuous signals on artifact-free segments before selecting NREM epochs.

At each frequency, phases were binned into 20 equal intervals between 0 and 2π, and the mean amplitude within each bin was normalized to form a phase-amplitude distribution. MI was computed as the Kullback-Leibler divergence from a uniform distribution [36]. Within-night surrogate distributions were generated by circularly time-shifting one signal relative to the other (60 seconds, which is ~ 3 EGG cycles, for gastric phase; 180 seconds, which is ~ 1.5 EGG ISO cycle, for the EGG ISO phase) and recomputing MI across 400 shuffles. For each channel and frequency, a signed z-score was computed as $z = \frac{MI_{emp} - \mu_{sur}}{\sigma_{sur}}$, where μ_{sur} and σ_{sur} are the surrogate mean and standard deviation, respectively.

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4.8 Statistical testing for modulation index spatial distributions

4.8.1 EGG power ISO phase and EEG sigma power ISO amplitude

To determined stage-dependent variations in EGG and EEG sigma ISO, we computed the difference in z-scored MI values between N2 and N3 within EEG sigma ISO frequencies (0.02 to 0.1 Hz) per channel. We ran a cluster-based permutation test (10,000 permutations; cluster-forming threshold set by the analytical t-critical value at α=0.05, two-sided). Channels belonging to clusters with $p < 0.05$ (FWER controlled) were highlighted on the topomap.

We averaged the N3 EGG-EEG sigma ISO MI z-scores during N3 MI within EEG sigma ISO frequencies (0.02 to 0.1 Hz) per channel and tested whether the group mean exceeded zero (one-sample, one-sided; Fig. 2g). A cluster-based permutation test (10,000 permutations; cluster-forming threshold set by the analytical t-critical value at α=0.05, one-sided). Channels belonging to clusters with $p < 0.05$ (FWER controlled) were highlighted on the topomap.

4.8.2 EGG phase and EEG amplitude at delta and sigma frequencies

For EEG-EGG MI during NREM, we used computed band-averaged values per channel for delta (0.5 to 4 Hz [40]) and sigma (9 to 16 Hz [34]) bands. MI z-scores were averaged across all frequency bins lying in the target band. Group-level significance on the scalp was assessed with a one-sample (band-averaged z-scored MI ¿ 0), one-sided cluster-based permutation test (10,000 permutations; cluster-forming threshold set at the analytical t-critical value at $\alpha = 0.05$, one-sided) [84]. Channels with clusters $p < 0.05$ were considered significant (FWER controlled) and highlighted on the topomap.

4.9 Event-triggered gastric amplitude modulation around cortical slow wave oscillations and spindles

We quantified gastric amplitude changes around cortical SWOs and SWO-spindle complexes using event-triggered analysis (ETA). Only events whose entire analysis window lay within the target sleep stage were retained, and uncoupled events were randomly subsampled to match the coupled counts per night.

EGG was filtered to the gastric band and analytic amplitude was computed via the Hilbert transform. For each retained event, we extracted a continuous epoch ± 40 seconds around the SWO negative peak and z-scored the EGG amplitude per epoch using the pre-event baseline (-40 to -10 s). Within each night, we averaged epochs across events to obtain coupled and uncoupled ETA curve.

For scalar summaries, we defined pre (-20 to 0 s) and post (0 to 20 s) windows, where 20 seconds corresponds to roughly one EGG cycle, and computed the absolute EGG amplitude modulation strength as $|\Delta z| = |\overline{z}_{post} - \overline{z}_{pre}|$ separately for coupled and uncoupled curves.

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4.10 Gastric phase concentration during cortical slow wave oscillations and spindles using the von Mises distribution

Instantaneous EGG phase was derived as described above. To quantify phase locking of the EGG, a von Mises distribution was fit and the concentration parameter, κ, [41] was obtained for every participant separately for coupled and uncoupled SWOs in N2 and N3. All fits required at least five events. Events were limited to complete NREM sleep cycles, and when uncoupled counts exceeded coupled counts, uncoupled events were count-matched to an average of 50 resamples. EGG κ values were determined at the SWO trough and during coupled spindle peaks, defined as a spindle that occurs after an SWO trough.

4.11 Linear mixed-effects statistical analyses

Linear mixed-effects models (LMMs) were implemented in Python using the statsmodels package to examine how gastric features varied across sleep cycles and sleep stages while accounting for repeated measurements, experimental condition, sex, and time since last meal. Each model included fixed effects for sleep stage (contrast-coded [**85**]) or sleep cycle (mean-centered, continuous), experimental condition (Adaptation, Control, or TMR; contrast-coded), sex (contrast-coded: male = -1, female = $+1$), and time since last meal (mean-centered, continuous), as well as a random intercept for experimental night to capture within-participant dependencies.

Because these models incorporate both within-night (Level 1) and between-night (Level 2) sources of variance, they are equivalent to multilevel (hierarchical) linear models that estimate population-level (fixed) effects while allowing subject-level (random) variability. The general model structure was:

$$y_{ij} = \beta_0 + \beta_1 X_{1,ij} + \beta_2 X_{2,ij} + \cdots + u_j + \epsilon_{ij},$$

where y_{ij} is the dependent variable of interest for night j and observation i, β_k are fixed-effect coefficients, $u_j \sim \mathcal{N}(0, \sigma_u^2)$ is the random intercept capturing between-night variance, and $\epsilon_{ij} \sim \mathcal{N}(0, \sigma^2)$ represents residual (within-night) error.

All models were fit using restricted maximum likelihood (REML) estimation with the L-BFGS optimizer, and significance was assessed using Walk z-statistics on the fixed-effect coefficients. Standardized effect sizes were computed as $d_{LMM} = \beta / \sqrt{\text{residual variance}}$.

4.11.1 Gastric power sleep-stage models

To quantify the main effect of sleep stage, we fit an LMM with fixed effects for sleep stage (contrast-coded NREM = +1, REM = -1), experimental condition (two orthogonal contrasts: Control - TMR and Experimental - Adaptation), sex (F = +1, M = -1), and time since last meal (centered relative to lights-off). The stage difference was summarized as $\Delta_{\text{NREM-REM}} = 2\beta_{\text{stage}}$ (Fig. 1d, Table 2).

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4.11.2 Gastric power cycle-dependent models

To examine gradual changes in gastric power across the night, we fit a separate LMM including fixed effects for sleep cycle (mean-centered), experimental condition, sex, and time since last meal referenced to the median onset of each sleep cycle. This covariate captures the postprandial interval at the start of each cycle and allowed separation of within-night temporal decline from meal-related effects (Fig. 1e, Table 1).

4.11.3 EGG infraslow oscillation power models

To assess stage-dependent differences in gastric ISO power (Fig. 2c, Fig. 7b), we fit a LMM with sleep stage (NREM, REM, and Wake) as a fixed factor, using NREM as the reference level, along with the previously used fixed effects and random intercept to account for within-participant dependencies.

Pairwise contrasts were computed between stages (NREM-REM, NREM-Wake, and REM-Wake) using the covariance of the fixed effects to obtain Wald z statistics and 95% confidence intervals. All reported significance values are two-sided and were adjusted for multiple comparisons across the three contrasts using the Benjamini-Hochberg false-discovery rate (FDR) correction ($q < 0.05$).

4.11.4 Phase-amplitude coupling (PAC) modulation index models

To test whether the strength of gastric-cortical infraslow coupling differed between N2 and N3, we fit LMMs to the MI values averaged over the 0.02 to 0.10 Hz range (Fig. 2f). The primary contrast of interest was sleep stage, contrast-coded as N2 = +1 and N3 = -1, such that the estimated difference was $\Delta_{\mathrm{N3-N2}} = -2\beta_{\mathrm{stage}}$.

ROI-specific models (Fig. 9b) used the same contrast structure and covariates, fit separately for frontal, central, parietal, temporal, and occipital regions. All significance values were derived from Wald z statistics and false-discovery rate (FDR) corrected (Benjamini-Hochberg, $q < 0.05$) across ROIs. Between-night variance estimates were near the boundary across models, indicating minimal residual variability after normalization.

4.11.5 Event-triggered models of gastric amplitude modulation

LMMs were used to test EGG amplitude modulation effects when aligned to the negative peak of SWOs and SWO-spindle complexes (see Fig. 3b, c). Two hierarchical mixed-effects models were used to capture within- and between-condition effects. The first, window model, tested EGG amplitude as a function of window (Pre vs. Post), coupling status (Coupled vs. Uncoupled), and their interaction. The second, delta model,) examined post-pre differences ($|\Delta z|$) as a function of coupling status. Both models included experimental condition, sex, and time since last meal as covariates.

Predictors were contrast-coded to yield interpretable coefficients: window ($+1$ = Pre, -1 = Post), coupling (-1 = Uncoupled, $+1$ = Coupled), and the same codes for the covariates as described above.

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4.11.6 EGG von Mises models

LMMs were used to test how κ varied with SWO coupling, stage, and sleep cycle while accounting for repeated nights. The primary model (Fig. 4c, d) included fixed effects for coupling status (contrast-coded: Uncoupled = -1, Coupled = $+1$), stage (N2 = -1, N3 = $+1$), their interaction, ROI, and the same codes for the covariates as described above.

Cycle-level models (Fig.9a) further examined how κ changed across consecutive sleep cycles within each stage, including fixed effects for coupling, cycle number (mean-centered), and their interaction, along with the same covariates. Region-specific contrasts (Fig. 9b, c) were tested using equivalent model structures applied separately within each ROI to estimate local coupling and stage effects. Between-night variance was small relative to fixed effects, indicating stable participant-level differences across sessions.

4.12 Sleep quality association analysis

4.12.1 Subjective sleep quality and objective quantity measures

Subjective sleep quality (SSQ) was assessed using the Schlaffragebogen-A (SF-A/R) questionnaire, which participants completed each morning after sleep [78]. The SF-A/R comprises 22 items probing perceived sleep depth, restfulness, continuity, dream quality, and morning well-being (refer to Table18 . Each item is rated on a 1–5 Likert scale (1 = "not at all", 5 = "very much"). Five negatively worded items (e.g., tired, exhausted, in need of sleep) were reverse-scored using the standard transformation $X' = 6 - X$. A composite SSQ value was then computed as the mean of all 22 items after reverse scoring, such that higher scores reflect better subjective sleep quality. The 22 SF-A/R items showed strong internal consistency (Cronbach's $\alpha = 0.85$), supporting the use of their mean as a composite subjective sleep quality score. To minimize confounds from experimental manipulation, all analyses were restricted to the Adaptation night.

Objective sleep quantity measures were derived from hypnograms and included total sleep time (TST; minutes), wake after sleep onset (WASO; minutes), and sleep maintenance efficiency (SME; ratio of TST to total time in bed).

For each participant, night, and NREM stage, we computed the von Mises concentration parameter (κ) of gastric phase during SWOs as described above. EGG κ values were averaged separately for coupled and uncoupled SWO events within each sleep stage (N2, N3). To normalize distributions, all models used the natural logarithm of κ ($\log \kappa$) as the dependent or independent variable.

4.12.2 Linear mixed-effects models

Associations between gastric phase alignment and sleep quality were tested using linear mixed-effects models (LMMs). Each model included a random intercept for participant to account for repeated measures, and fixed effects for condition contrasts, sex, and time since last meal as described above. Continuous predictors were z-scored prior to fitting.

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For the SSQ analysis (Fig. 4d), we examined whether the relationship between sleep quality and gastric phase concentration differed by coupling status during N3 SWOs. The model included an interaction between SSQ score and coupling. Wald z-statistics were used to test fixed effects. Per-coupling simple slopes were computed using linear contrasts of the fitted model, with p-values adjusted using the Benjamini–Hochberg false discovery rate (FDR) correction across slopes.

For the summary models shown in Supplementary Fig. 10a,b, we repeated these analyses across all NREM stage × coupling conditions (N2 coupled SWOs, N2 uncoupled SWOs, N3 coupled SWOs, N3 uncoupled SWOs). In the subjective analyses, sleep quality score was treated as the dependent variable, and z-scored $\log \kappa$ served as the predictor (Fig. 10a). The resulting standardized coefficients (β_1) represent the change in sleep quality per standard deviation increase in gastric phase concentration. The same model form was applied to objective outcomes (SME, TST, WASO), each entered as the dependent variable, with p-values FDR-corrected within outcome across the four stage × coupling tests (Fig. 10b). All tests were two-sided with $\alpha = 0.05$. Reported estimates reflect fixed effects ($\beta \pm \text{SE}$), Wald 95% confidence intervals, and FDR-corrected significance levels.

4.12.3 Explained variance and autonomic feature analyses

To assess whether stomach-brain coupling and intrinsic gastric dynamics explain unique variance in perceived sleep quality beyond canonical PSG metrics, we fit a series of hierarchical ordinary least squares (OLS) regression models using data from the Adaptation night.

The baseline PSG model included macro-architectural sleep composition (%N1, %N2, %N3, %Wake), indices of sleep fragmentation (sleep to wake and N3 to N1/N2 transitions per hour), and covariates for sex, experimental condition, and time since last meal. EGG predictors were then added in successive models to test their incremental explanatory power.

First, we added the EGG κ term (log-transformed) derived from N3 uncoupled events, which showed the strongest correspondence with SSQ. Next, we quantified intrinsic gastric slow-wave features reflecting autonomic tone independent of cortical coupling, including total EGG power (0.01 to 0.15 Hz), the percentage of brady-, normo-, and tachygastric states, and the infraslow (0.005 to 0.01 Hz) amplitude variance of the EGG envelope. Parallel analyses incorporated cardiac autonomic measures derived from the ECG, including mean heart rate (HR), heart-rate variability (HRV; RMSSD), and HR infraslow power (0.02 to 0.07 Hz).

Nested model comparisons were evaluated using partial F-tests, and incremental effect sizes were quantified by ΔR^2, partial R^2, and Cohen's f^2. Five-fold cross-validation was used to estimate out-of-sample performance. Standardized β coefficients were computed after z-scoring all predictors to allow effect-size comparability across terms.

Supplementary information. Supplementary figures and tables can be found in the subsequent section.

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Declarations

If any of the sections are not relevant to your manuscript, please include the heading and write 'Not applicable' for that section.

5 Supplementary Material

Table 1 Linear mixed-effects model predicting gastric power across sleep cycles. LMM contrast-coded and controlling for sex, condition, and time since last meal (adjusted to sleep cycle start time).

Predictor	**Coef.**	**Std. Err.**	z	$P > \|z\|$	**95% CI**
Intercept	0.9960	0.0049	204.6820	0.0000	[0.9866, 1.0056]
Sleep cycle	-0.0270	0.0029	-9.1084	0.0000	[-0.0327, -0.0211]
Condition (Control–TMR)	0.0020	0.0054	0.4120	0.6803	[-0.0083, 0.0128]
Condition (Exp.–Adaptation)	0.0010	0.0048	0.2010	0.8407	[-0.0085, 0.0104]
Time since meal (h, centered)	-0.0030	0.0011	-2.7659	0.0057	[-0.0054, -0.0009]
Sex (F=+1, M=-1)	0.0040	0.0044	0.9964	0.3190	[-0.0042, 0.0129]
Group Var (participant random intercept)[1]	0.0000				

[1]Estimated at boundary (≈ 0), indicating negligible between-participant variance after normalization.

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Table 2 Linear mixed-effects model predicting normalized gastric power (stage). LMM for normalized gastric power with contrast-coded sleep stage and covariates (random intercepts by participant).

Predictor	**Coef.**	**Std. Err.**	z	$P > \|z\|$	**95% CI**
Intercept	1.0000	0.0220	45.3760	0.0000	[0.9568, 1.0431]
Sleep stage (NREM–REM, contrast-coded ±1)	0.1190	0.0162	7.3187	0.0000	[0.0870, 0.1507]
Condition (Control–TMR)	-0.0000	0.0244	-0.0000	1.0000	[-0.0478, 0.0478]
Condition (Exp.–Adaptation)	-0.0000	0.0227	-0.0000	1.0000	[-0.0444, 0.0444]
Time since meal (h, centered)	0.0000	0.0058	0.0000	1.0000	[-0.0113, 0.0113]
Sex (F–M)	0.0000	0.0206	0.0000	1.0000	[-0.0404, 0.0404]
Group Var (participant random intercept)[1]	0.0000				

[1]Estimated at boundary (≈ 0), indicating negligible between-participant variance after normalization.

Table 3 Linear mixed-effects model predicting gastric power with sleep stage × sleep cycle. LMM contrast-coded and controlling for sex, condition, and time since last meal.

Predictor	**Coef.**	**Std. Err.**	z	$P > \|z\|$	**95% CI**
Intercept	1.1150	0.0110	100.6360	0.0000	[1.0930, 1.1360]
Sleep stage (REM–NREM)	-0.0900	0.0260	-3.5380	0.0004	[-0.1410, -0.0400]
Sleep cycle (2–1)	-0.0970	0.0120	-8.2910	0.0000	[-0.1200, -0.0740]
Sleep cycle (3–1)	-0.1350	0.0120	-11.3790	0.0000	[-0.1590, -0.1120]
Sleep cycle (4–1)	-0.1460	0.0130	-11.5310	0.0000	[-0.1710, -0.1210]
Sleep cycle (5–1)	-0.1350	0.0170	-7.9980	0.0000	[-0.1680, -0.1020]
Condition (Control–Adapt.)	-0.0020	0.0090	-0.2420	0.8088	[-0.0200, 0.0160]
Condition (TMR–Adapt.)	-0.0040	0.0080	-0.4710	0.6379	[-0.0200, 0.0120]
Sex (F=+1, M=-1)	-0.0060	0.0080	-0.6800	0.4964	[-0.0220, 0.0110]
REM × Cycle 2	0.0790	0.0320	2.4880	0.0128	[0.0170, 0.1420]
REM × Cycle 3	0.0480	0.0330	1.4600	0.1443	[-0.0170, 0.1130]
REM × Cycle 4	0.0830	0.0320	2.6180	0.0088	[0.0210, 0.1440]
REM × Cycle 5	0.1000	0.0380	2.6000	0.0093	[0.0250, 0.1760]
Time since meal (h, centered)	-0.0020	0.0010	-1.4710	0.1413	[-0.0040, 0.0010]
Group Var (participant random intercept)[1]	0.0000				

[1]Estimated at boundary (≈ 0), indicating negligible between-participant variance.

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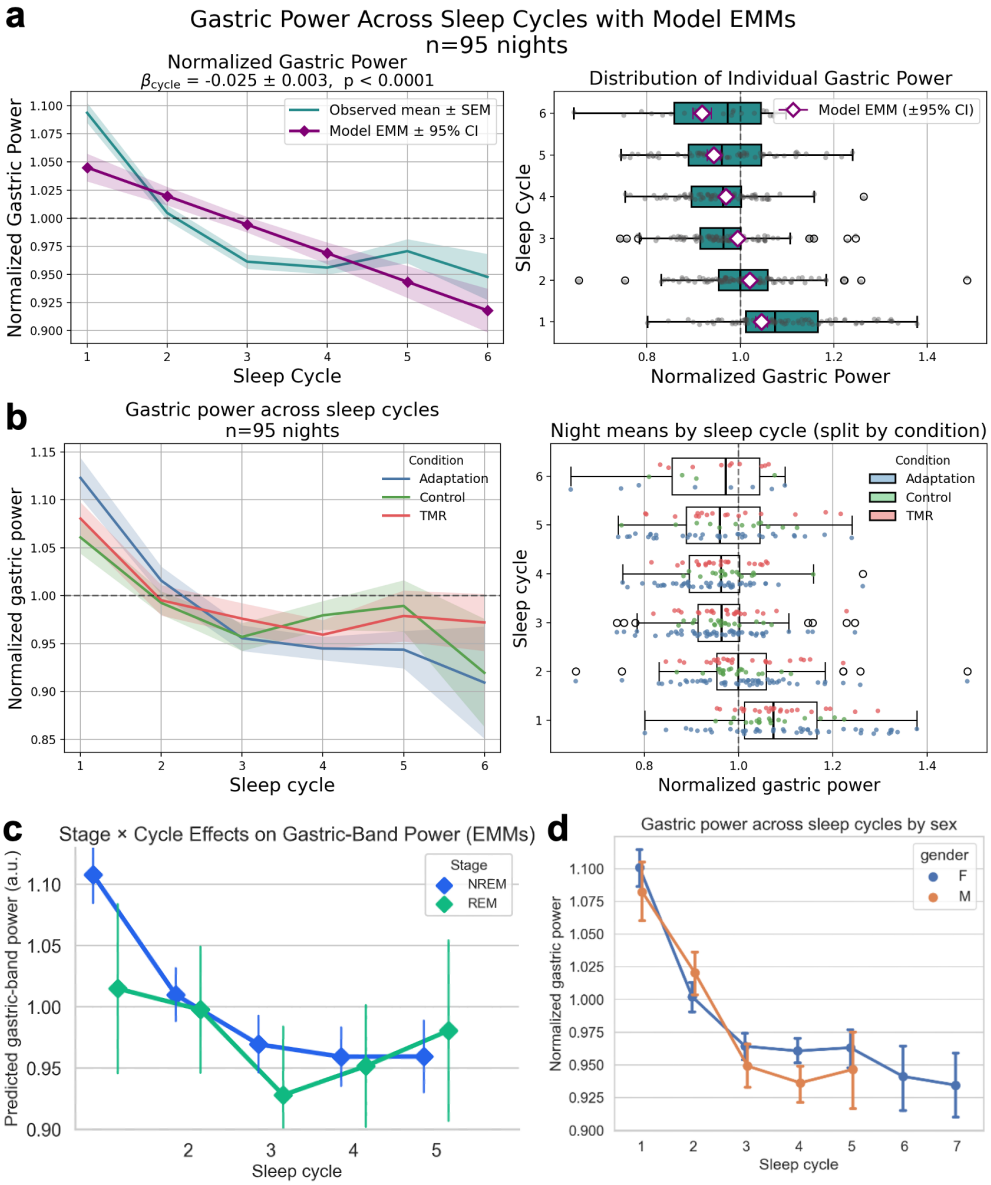


Fig. 6 **(a)** Visualization of normalized gastric slow-wave power across successive sleep cycles (n = 95 nights). Left: Observed means ± SEM (teal) and model-estimated marginal means (EMMs) ± 95 % CI (magenta), derived from a linear mixed-effects model controlling for sex, condition, and time since last meal ($\beta_{cycle} = -0.027 \pm 0.003$, $p < 0.0001$; n = 95 nights). Right: Night-level distributions of normalized gastric power per cycle, with model EMMs (diamonds) and ± 95 % CI bars. **(b)** Same visualization as in (a), but split by experimental condition (Adaptation = blue, Control = green, Targeted Memory Reactivation (TMR) = red), showing a similar downward trend across all conditions. **(c)** Predicted gastric power EMMs across cycles separately for NREM and REM sleep, indicating a stable NREM ¿ REM difference with no significant stage × cycle interaction. **(d)** Gastric power trajectories by sex (female = blue, male = orange), demonstrating comparable night-to-night declines in both groups.

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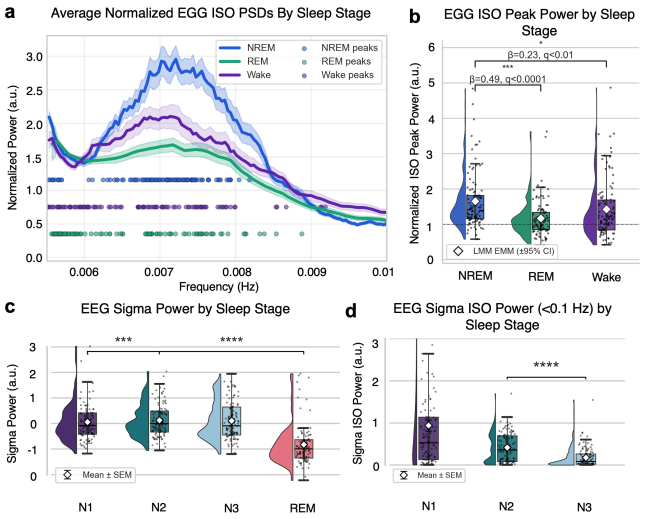


Fig. 7 **(a)** Group-level average normalized EGG infraslow (ISO; 0.005–0.010 Hz) power spectra across sleep–wake stages (n = 107 nights). Mean ± SEM (solid lines ± shaded area) and individual peak frequencies (dots) are shown, revealing strongest ISO power during NREM sleep. **(b)** Violin plots of normalized ISO peak power by stage (n = 102 nights), with linear mixed-effects model (LMM) estimated marginal means (diamonds ± 95 % CI). ISO power was significantly higher in NREM than in REM ($\beta = 0.23$, $p_{LMM} < 0.01$, corrected) and Wake ($\beta = 0.49$, $p_{LMM} < 0.0001$, corrected). **(c)** Stage-dependent EEG sigma-band power (z-scored within night; $n = 96$ nights). Raincloud plots show per-night distributions, individual values, and mean ± SEM. Significantly lower sigma power in N1 compared with N2 ($\beta = -0.089$, $p_{LMM} < 0.001$) was observed, with no significant difference between N3 and N2 ($p_{LMM} = 0.48$), and markedly reduced sigma power during REM relative to N2 ($\beta = -0.960$, $p_{LMM} < 0.0001$). **(d)** Stage-dependent EEG sigma-band infraslow (ISO; 0.01–0.1 Hz) power across NREM stages (n = 107 nights). Raincloud plots show per-night distributions (violin), individual values (points), and mean ± SEM (white diamonds). A linear mixed-effects model revealed significantly lower sigma ISO power in N3 relative to N2 ($\beta = -0.67$, $p_{LMM} < 0.001$), while N1 did not differ from N2 ($p_{LMM} = 0.55$), indicating that sigma-band ISOs remain reliably present during N3 despite reduced amplitude.

Table 4 Linear mixed-effects model predicting z-scored sigma power. Fixed effects included sleep stage (N2 reference), experimental condition (Adaptation reference), sex (contrast-coded: M = −1, F = +1), and mean-centered time since meal. Random intercepts were estimated per subject-session (night).

Predictor	Coef.	Std. Err.	z	$P > \|z\|$	95% CI
Intercept	-0.008	0.075	-0.100	0.920	[-0.155, 0.140]
N1 – N2	-0.089	0.022	-4.038	0.000	[-0.133, -0.046]
N3 – N2	-0.016	0.022	-0.707	0.480	[-0.059, 0.028]
REM – N2	-0.960	0.022	-43.297	0.000	[-1.004, -0.917]
Control – Adaptation	0.120	0.128	0.930	0.352	[-0.132, 0.371]
TMR – Adaptation	0.368	0.118	3.122	0.002	[0.137, 0.600]
Sex (F vs. M)	0.040	0.060	0.661	0.508	[-0.078, 0.158]
Time since meal (z)	-0.017	0.051	-0.336	0.737	[-0.116, 0.082]
Group variance	0.217	0.058			

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Table 5 Linear mixed-effects model predicting sigma ISO power. LMM predicting log-transformed EEG sigma-band ISO power (0.01–0.1 Hz). The reference level for stage is N2 and for condition is Adaptation. Random intercepts were fit per night.

Predictor	Coef.	Std. Err.	z	$P > \|z\|$	95% CI
Intercept	-0.582	0.162	-3.588	< 0.001	[-0.899, -0.264]
N1 vs. N2	0.127	0.213	0.593	0.553	[-0.292, 0.545]
N3 vs. N2	-0.665	0.165	-4.032	< 0.001	[-0.988, -0.342]
Control vs. Adaptation	-0.122	0.184	-0.663	0.508	[-0.484, 0.239]
TMR vs. Adaptation	-0.175	0.170	-1.030	0.303	[-0.508, 0.158]
Sex (F=+1, M=-1)	0.007	0.087	0.077	0.938	[-0.163, 0.177]
Time since meal (z)	0.028	0.074	0.381	0.703	[-0.117, 0.174]
Samples per stage (z)	-0.034	0.091	-0.370	0.711	[-0.211, 0.144]
Night intercept variance	0.356	0.150			

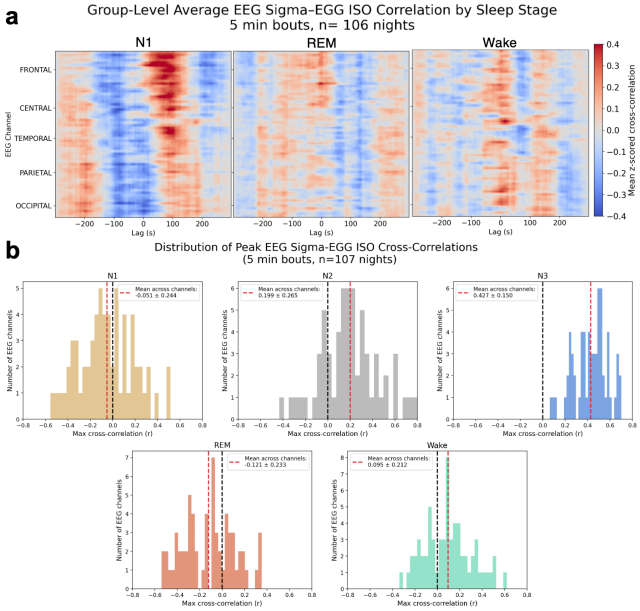


Fig. 8 **(a)** Group-level mean z-scored cross-correlations between EEG sigma-band power and EGG ISO amplitude, computed in 5-min bouts (n = 106 nights). Red bands indicate positive coupling at maximal lags. **(b)** Distributions of peak EEG sigma-EGG ISO cross-correlation values across EEG channels for each sleep stage, with mean ± SD indicated (vertical red line ± value). Strongest positive correlations were observed during N3 sleep, consistent with enhanced gastric-sigma synchrony during consolidated sleep.

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Table 6 ISO Cross-correlations by ROI and NREM sleep stage. Group-level averages of maximal EGG–EEG sigma ISO cross-correlation values during N2 and N3 sleep.

Cortical Region	Maximal correlation (group mean, z-scored)	
	N2	N3
Frontal	0.332 ± 0.227	0.469 ± 0.165
Central	0.075 ± 0.184	0.366 ± 0.197
Parietal	-0.013 ± 0.236	0.408 ± 0.106
Occipital	0.266 ± 0.175	0.444 ± 0.106
Temporal	0.429 ± 0.211	0.457 ± 0.129
All	0.0199 ± 0.265	0.427 ± 0.150

Table 7 LMM predicting EEG-EGG ISO MI. Linear mixed-effects model predicting ISO modulation index (MI) in parietal regions. LMM contrast-coded and controlling for sex, condition, and time since last meal.

Predictor	Coef.	Std. Err.	z	$P > \|z\|$	95% CI
Intercept	0.246	0.085	2.88	0.004	[0.078, 0.413]
Sleep stage (N2=+1, N3=-1)	-0.116	0.054	-2.14	0.033	[-0.222, -0.010]
Condition (Control–TMR)	0.278	0.087	3.18	0.001	[0.107, 0.450]
Condition ([(Control+TMR)/2]–Adapt.)	0.402	0.094	4.28	0.000	[0.218, 0.587]
Time since meal (centered)	-0.021	0.019	-1.08	0.279	[-0.059, 0.017]
Sex (F=+1, M=-1)	0.081	0.078	1.04	0.299	[-0.072, 0.234]
Group Var (participant random intercept)[1]	0.162	0.142	1.14	0.253	[-0.116, 0.441]

[1]Estimated at boundary (≈ 0) indicates negligible between-participant variance after normalization.

Table 8 Linear mixed-effects model predicting ROI-wise ISO modulation index. LMM includes fixed effects for condition, sex, and time since last meal (stage-specific), and a random intercept per night.

ROI	n	Coef.$_{\text{stage}}$	Δ (N3–N2)	95% CI	z	p_{one}	d_{LMM}
Central	90	-0.045	0.090	[-0.112, 0.293]	-0.874	0.191	0.139
Frontal	90	-0.051	0.101	[-0.082, 0.284]	-1.082	0.140	0.171
Occipital	90	-0.098	0.196	[-0.049, 0.440]	-1.568	0.058	0.247
Parietal	90	-0.117	0.234	[0.025, 0.442]	-2.192	0.014	0.351
Temporal	90	-0.115	0.231	[0.022, 0.439]	-2.168	0.015	0.347

Group variance estimated at boundary (≈ 0).

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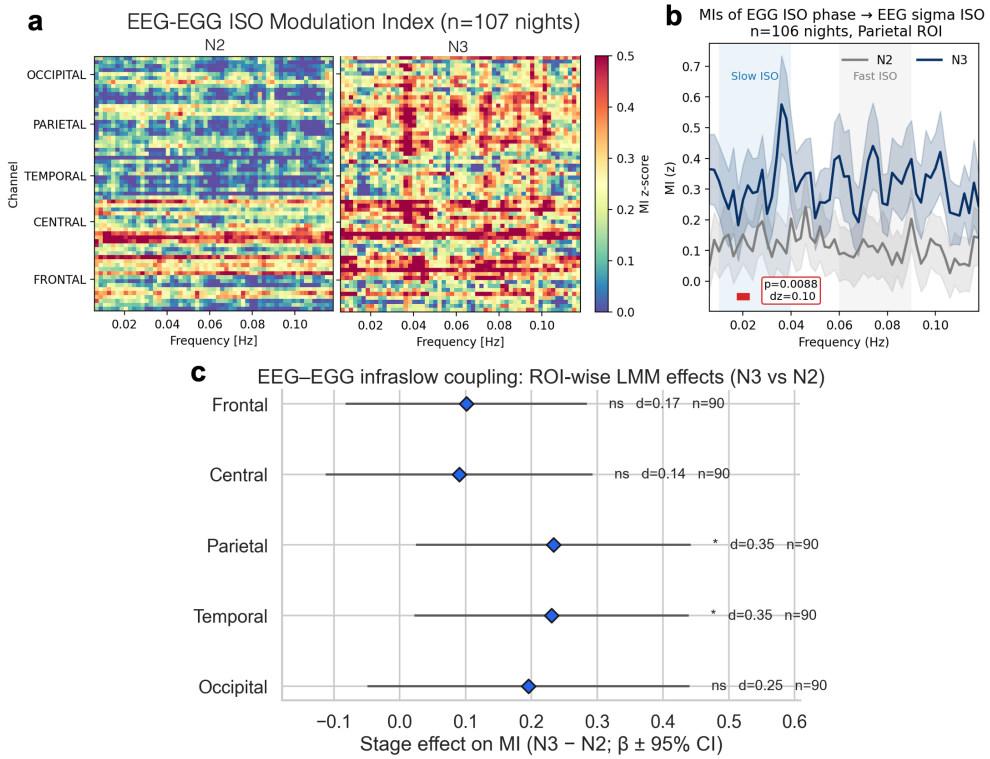


Fig. 9 **(a)** Visualization of z-scored modulation index between the phase of the EGG power infraslow oscillation (ISO) and amplitude of the EEG sigma ISO from 0.02 to 0.1 Hz during N2 (left) and N3 (right) sleep stages (n = 107 nights). **(b)** A cluster-based permutation test further identified a significant frequency-specific enhancement centered at 0.02 Hz ($p = 0.0088$, corrected; $d_z = 0.10$). **(c)** Forest plot shows linear mixed-effects estimates ($\beta \pm 95\%$ CI) of the N3–N2 difference in EGG-EEG ISO PAC per region of interest (ROI), controlling for repeated measures per night; stars reflect FDR-adjusted significance across ROIs. Standardized effect size from the LMM and the number of paired nights contributing to each ROI are reported.

Table 9 Linear mixed-effects model predicting post-pre EGG amplitude difference. LMM of post–pre EGG amplitude differences ($\Delta_{\text{Post–Pre}}$) as a function of coupling status, with covariate adjustment.

Predictor	Coef.	Std. Err.	z	$P > \|z\|$	95% CI
Intercept (grand mean)	0.290	0.039	7.398	< 0.001	[0.214, 0.367]
Coupling (Coupled–Uncoup)	0.062	0.004	14.339	< 0.001	[0.054, 0.071]
Condition (Control vs. TMR)	0.038	0.033	1.153	0.249	[-0.027, 0.103]
Condition (Adaptation vs. Others)	0.038	0.030	1.290	0.197	[-0.020, 0.097]
Sex (F–M)	-0.006	0.037	-0.172	0.863	[-0.079, 0.066]
Time since meal (h, centered)	-0.001	0.009	-0.116	0.907	[-0.018, 0.016]

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Table 10 Linear mixed-effects model predicting absolute EGG amplitude (window × coupling). LMM for absolute EGG amplitude ($|z|$) during NREM sleep, including main and interaction effects of time window (Pre vs. Post) and coupling status (Coupled vs. Uncoupled). Also includes fixed effects for condition, sex, and time since last meal (stage-specific), and a random intercept per night.

Predictor	Coef.	Std. Err.	z	$P > \|z\|$	95% CI
Intercept (grand mean)	0.261	0.026	10.155	< 0.001	[0.211, 0.311]
Window (Post–Pre)	-0.129	0.003	-46.761	< 0.001	[-0.134, -0.124]
Coupling (Coupled–Uncoup)	0.065	0.003	23.407	< 0.001	[0.059, 0.070]
Window × Coupling	-0.031	0.003	-11.276	< 0.001	[-0.037, -0.026]
Condition (Control vs. TMR)	0.028	0.028	1.024	0.306	[-0.026, 0.083]
Condition (Adaptation vs. Others)	0.025	0.025	0.998	0.318	[-0.024, 0.073]
Sex (F–M)	-0.004	0.024	-0.171	0.864	[-0.051, 0.043]
Time since meal (h, centered)	-0.006	0.007	-0.921	0.357	[-0.020, 0.007]

Table 11 Linear mixed-effects model results for von Mises κ during N3 sleep. Fixed effects include coupling, sleep cycle, their interaction, experimental condition contrasts, time since last meal (centered), and sex (F=+1, M=-1). Random intercepts for participant and a night variance component.

Predictor	Coef.	Std. Err.	z	$P > \|z\|$	95% CI
Intercept	-0.737	0.099	-7.431	< 0.001	[-0.932, -0.543]
Coupling (centered)	0.409	0.035	11.548	< 0.001	[0.340, 0.478]
Sleep cycle (centered)	0.143	0.032	4.506	< 0.001	[0.081, 0.206]
Coupling × Cycle	-0.009	0.028	-0.319	0.750	[-0.064, 0.046]
Condition contrast 1	-0.118	0.072	-1.633	0.102	[-0.259, 0.024]
Time since meal (centered, h)	0.043	0.019	2.317	0.021	[0.007, 0.079]
Condition contrast 2	0.030	0.075	0.405	0.685	[-0.116, 0.177]
Sex (F=+1, M=-1)	-0.201	0.092	-2.192	0.028	[-0.380, -0.021]
Participant Var	0.325	0.127	2.553	0.011	[0.076, 0.575]
Night Var	0.042	0.081	0.515	0.607	[-0.117, 0.200]

Table 12 Linear mixed-effects model results for von Mises κ during N2 sleep. Same fixed and random effects as Table 11.

Predictor	Coef.	Std. Err.	z	$P > \|z\|$	95% CI
Intercept	-1.044	0.068	-15.302	< 0.001	[-1.177, -0.910]
Coupling (centered)	0.202	0.030	6.726	< 0.001	[0.143, 0.261]
Sleep cycle (centered)	-0.053	0.025	-2.117	0.034	[-0.102, -0.004]
Coupling × Cycle	-0.005	0.023	-0.224	0.823	[-0.051, 0.041]
Condition contrast 1	0.015	0.074	0.202	0.840	[-0.131, 0.161]
Time since meal (centered, h)	-0.016	0.017	-0.946	0.344	[-0.050, 0.018]
Condition contrast 2	0.027	0.078	0.345	0.730	[-0.126, 0.180]
Sex (F=+1, M=-1)	-0.094	0.062	-1.517	0.129	[-0.214, 0.027]
Participant Var	0.000	0.058	0.000	1.000	[-0.113, 0.113]
Night Var	0.185	0.078	2.372	0.018	[0.032, 0.337]

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Table 13 Full linear mixed-effects model predicting gastric phase concentration during cortical SWOs. LMM predicting log(κ) as a function of coupling (Coupled vs. Uncoupled), sleep stage (N2 vs. N3), and cortical region of interest (ROI), including their interactions and covariates for experimental condition, sex, and time since last meal. Random intercepts were included per participant (REML estimation, L-BFGS optimizer). Results correspond to the model summarized in Fig. 4d and Fig.9.

Predictor	**Estimate (β)**	**Std. Err.**	z	$P > \|z\|$	**95% CI**
Intercept	-1.435	0.077	-18.58	< 0.001	[–1.586, –1.283]
ROI [Central]	0.170	0.061	2.77	0.006	[0.050, 0.290]
ROI [Posterior]	-0.135	0.059	-2.28	0.022	[–0.250, –0.019]
Coupling (C)	0.540	0.042	12.83	< 0.001	[0.457, 0.622]
C × ROI [Central]	0.078	0.061	1.27	0.204	[–0.042, 0.198]
C × ROI [Posterior]	-0.029	0.059	-0.49	0.628	[–0.144, 0.087]
Stage (N3–N2)	0.094	0.042	2.23	0.026	[0.011, 0.177]
Stage × ROI [Central]	-0.038	0.061	-0.62	0.536	[–0.158, 0.082]
Stage × ROI [Posterior]	0.029	0.059	0.50	0.620	[–0.086, 0.145]
C × Stage	0.071	0.042	1.68	0.093	[–0.012, 0.153]
C × Stage × ROI [Central]	0.088	0.061	1.43	0.153	[–0.032, 0.208]
C × Stage × ROI [Posterior]	0.063	0.059	1.06	0.289	[–0.053, 0.178]
Condition [Contrast 1]	0.070	0.044	1.58	0.115	[–0.017, 0.156]
Condition [Contrast 2]	0.114	0.040	2.83	0.005	[0.035, 0.193]
Sex (F–M)	-0.121	0.066	-1.82	0.068	[–0.251, 0.009]
Time since meal (h)	-0.006	0.012	-0.53	0.595	[–0.029, 0.017]

Table 14 Linear mixed-effects models relating gastric phase concentration (κ) to objective and subjective sleep quality measures. Objective sleep quality measures include SME: sleep maintenance efficiency; TST: total sleep time; and WASO: wake after sleep onset.

Outcome	Stage	Coupling	β_{std}	SE	z	p	q	n
SME	N2	Coupled	-3.4394	1.6577	-2.0748	0.038	0.076	49
SME	N2	Uncoupled	-0.7415	1.4393	-0.5152	0.606	0.809	49
SME	N3	Coupled	-0.0106	1.4186	-0.0074	0.994	0.994	46
SME	N3	Uncoupled	-7.8240	1.5191	-5.1504	2.6e-07	1.04e-06	48
TST	N2	Coupled	-9.2433	11.2229	-0.8236	0.410	0.820	49
TST	N2	Uncoupled	-5.4415	11.6489	-0.4671	0.640	0.854	49
TST	N3	Coupled	-0.8790	11.1959	-0.0785	0.937	0.937	46
TST	N3	Uncoupled	-27.3636	9.3950	-2.9126	0.00358	0.0143	48
WASO	N2	Coupled	13.3929	3.5268	3.7975	0.000146	0.000292	49
WASO	N2	Uncoupled	3.2214	5.9721	0.5394	0.590	0.766	49
WASO	N3	Coupled	-0.7252	2.4405	-0.2972	0.766	0.766	46
WASO	N3	Uncoupled	28.0566	1.0035	27.9581	5.26e-172	2.1e-171	48
sleep_quality_score	N2	Coupled	-0.0162	0.0742	-0.2185	0.827	0.874	49

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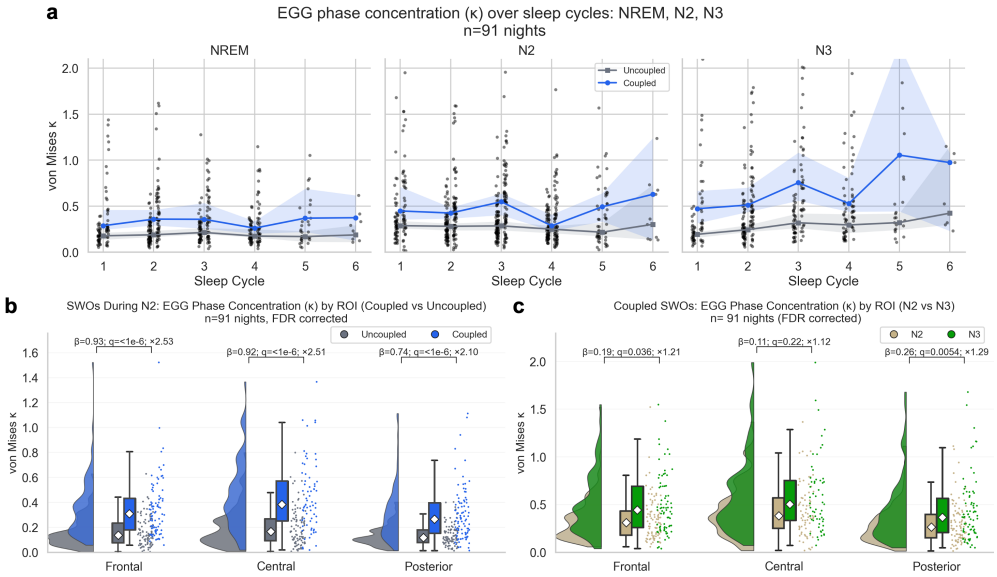


Fig. 10 **(a)** EGG phase concentration (κ) across sleep cycles for NREM, N2, and N3 (n = 91 nights). Linear mixed-effects models revealed a significant main effect of coupling (all $p_{LMM} < 10^{-6}$, corrected) and a progressive increase in coupling strength across cycles during N3 ($\beta = 0.110.04$, $p_{LMM} = 0.018$). **(b)** During N2, coupled SWO events exhibited higher κ than uncoupled events across all cortical regions (frontal: $\beta = 0.93$; central: $\beta = 0.92$; posterior: $\beta = 0.76$; all $p_{LMM} \leq 10^{-6}$, corrected; 2.1 to 2.5 fold increase). **(c)** Within coupled SWOs, EGG κ values increased during N3 compared to N2, specifically in frontal ($\beta = 0.19$, $p_{LMM} = 0.036$ corrected, 1.21 fold increase, n = 91 nights) and posterior regions ($\beta = 0.26$, $p_{LMM} = 0.0054$ corrected, 1.29 fold increase, n = 91 nights).

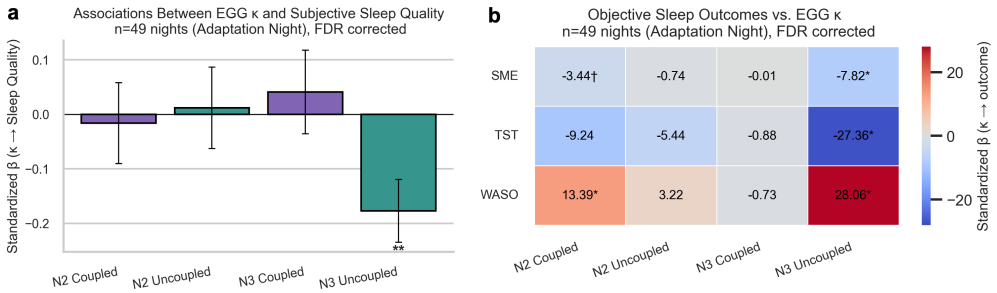


Fig. 11 **(a)** Standardized regression coefficients (β) from linear mixed-effects models relating EGG κ values to subjective sleep quality across NREM stage × coupling conditions during the adaptation night (n = 49 nights). Only uncoupled N3 κ showed a significant negative association with self-reported sleep quality ($\beta = -0.22 \pm 0.07$, corrected $p_{LMM} < 0.01$). **(b)** Standardized β coefficients linking EGG κ to objective sleep outcomes: sleep maintenance efficiency (SME), total sleep time (TST), and wake after sleep onset (WASO), with FDR correction across the 4 stage × coupling conditions. Higher κ during uncoupled N3 predicted longer WASO and shorter TST.

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Table 15 PSG-derived measures + EGG κ subjective sleep quality model. Standardized coefficients from PSG-only and PSG + κ models predicting subjective sleep quality. Adding log-transformed EGG κ did not improve explained variance ($\Delta R^2 = 0.0005$, partial $R^2 = 0.0005$, $p = 0.84$), and PSG coefficients remained virtually unchanged.

Predictor	**Standardized $\beta_{\text{PSG-only}}$**	**Standardized $\beta_{\text{PSG}+\kappa}$**	$p_{\text{PSG}+\kappa}$
Intercept	3.07	3.07	$< 1 \times 10^{-60}$
% N1	-0.112	-0.112	0.098
% N2	-0.222	-0.222	0.017
% N3	-0.200	-0.200	0.023
% Wake	-0.231	-0.231	0.010
Sleep→Wake (/h)	-0.102	-0.102	0.195
N3→N1/N2 (/h)	-0.083	-0.083	0.160
Coupling status (0/1)	0.004	0.004	0.94
Condition (contrast 1)	-0.033	-0.033	0.12
Condition (contrast 2)	0.023	0.023	0.36
Sex (contrast-coded)	-0.045	-0.045	0.46
Time since meal (h)	0.079	0.079	0.16
EGG $\log \kappa$	—	-0.012	0.84

Table 16 PSG + autonomic predictors to assess subjective sleep quality. Incremental variance explained by autonomic predictors of subjective sleep quality (Adaptation night). Each model adds the listed autonomic feature(s) to a baseline model containing PSG macro-architectural and fragmentation metrics.

Model	**Added predictor(s)**	R^2_{base}	R^2_{full}	ΔR^2	**partial R^2**	p **(F-test)**
M1	PSG-only	0.229	0.229	–	–	–
M2	+ EGG ISO amp SD (0.005–0.01 Hz)	0.229	0.356	0.127	0.165	0.014
M3	+ HRV RMSSD (ms)	0.229	0.250	0.021	0.026	0.345
M4	+ HR mean (bpm)	0.229	0.255	0.026	0.034	0.286
M5	+ HR infraslow power (0.02–0.07 Hz)	0.229	0.239	0.010	0.013	0.510
M6	+ EGG ISO + HRV RMSSD	0.229	0.367	0.138	0.179	0.039
M7	+ EGG ISO + HR infraslow	0.229	0.366	0.136	0.177	0.040
M8	+ EGG κ (log)	0.229	0.230	0.001	0.001	0.842

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Table 17 Partial correlations and incremental variance explained in subjective sleep quality (SSQ). Partial correlations show associations between each physiological feature and SSQ after controlling for PSG covariates. Hierarchical model comparisons report the incremental variance explained (ΔR^2) when adding each feature to the baseline PSG model.

Feature / Model	**Partial** r	p	ΔR^2	**Partial** R^2	p_{model}	**Adj.** R^2
Partial correlations (PSG-controlled)						
EGG ISO amp SD (0.005-0.01 Hz)	0.39	0.0088	–	–	–	–
EEG sigma ISO amp SD (0.005-0.01 Hz)	-0.37	0.016	–	–	–	–
Parietal MI (EGG-EEG ISO coupling)	0.24	0.12	–	–	–	–
Incremental explained variance (ΔR^2 comparisons)						
PSG-only baseline	–	–	0.0000	–	–	0.01
+ EGG ISO amp SD	–	–	0.13	0.16	0.019	0.14
+ EEG sigma ISO amp SD	–	–	0.11	0.13	0.031	0.11
+ Parietal MI (EGG-EEG coupling)	–	–	0.05	0.06	0.17	0.04
+ EGG ISO + EEG sigma ISO	–	–	0.23	0.28	0.0047	0.25

Table 18 SF-A/R items used to compute the subjective sleep quality (SSQ) score. Reverse-scored items indicated in parentheses.

Question Number	**Item wording**
1	I slept restfully
2	I slept well
3	I slept deeply
4	I woke up feeling refreshed
5	I woke up feeling rested
6	I fell asleep easily
7	I woke up easily
8	I had uninterrupted sleep
9	I slept without restlessness
10	I felt awake
11	I felt able to concentrate
12	I felt balanced
13	I felt fit
14	I felt carefree
15	I felt good about my dreams
16	I did not have disturbing dreams
17	My dreams were pleasant
18	I felt dozy (reverse-scored)
19	I felt exhausted (reverse-scored)
20	I felt in need of sleep (reverse-scored)
21	I felt tired (reverse-scored)
22	I felt unable to cope (reverse-scored)

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